Regulations and Curriculum for

Master of Computer Applications (MCA)

Version 2022.05

NITTE
(Deemed to be University)

(Established under Section 3 of UGC Act, 1956)
Placed under Category 'A' by MHRD, GoI | Accredited with 'A+' Grade by NAAC

Regulations and Curriculum for

Master of Computer Applications (MCA)

Choice Based Credit System (CBCS)

Applicable to the batch admitted from AY 2024-25 and onwards

Effective from AY 2024-25

Deemed to be University under Section 3 of UGC Act, 1956)
(Placed under Category 'A' by MHRD, Govt. of India, Accredited with 'A+' Grade by NAAC)
University Enclave, Medical Sciences Complex, Deralakatte,
Mangaluru – 575 018, Karnataka INDIA
Tel: +91-824-2204300/01/02/03, Fax: 91-824-2204305
Website: www.nitte.edu.in E-mail: info@nitte.edu.in

VISION

To build a humane society through excellence in the education and healthcare

MISSION

To develop
Nitte (Deemed to be University)
As a centre of excellence imparting quality education,
Generating competent, skilled manpower to face the scientific and social
challenges with a high degree of credibility, integrity,
ethical standards and social concern

NMAM INSTITUTE OF TECHNOLOGY

off-campus Institution of Nitte (Deemed to be University)
NITTE-574110, Karkala Taluk, Udupi District, Karnataka, India ISO 9001:2015 Certified, Accredited by NAAC with "A" Grade

Vision Statement

Pursuing Excellence, Empowering people, Partnering in Community Development

Mission Statement

To develop N.M.A.M. Institute of Technology, Nitte, as Centre of Excellence by imparting Quality Education to generate competent,
Skilled and Humane Manpower to face emerging Scientific, Technological,
Managerial and Social Challenges
with Credibility, Integrity, Ethics and Social Concern.

MCA DEPARTMENT VISION

Equipping students with computing and programming domain expertise with the state of the art technology solutions to enable them to meet global professional challenges.

MCA DEPARTMENT MISSION

The department strives to create an environment conducive to equipping students with teamwork ability, Professional Ethics. Sound Technical Knowledge and Skills to Handle Technological Challenges.

MCA SYLLABUS

Effective from

Academic Year

2024 – 2025

Applicable to the batch admitted from AY 2024-25 and onwards

Version 2022.05

With Scheme of Teaching & Examination

REGULATIONS: 2022.05
for
MCA Program
CHOICE BASED CREDIT SYSTEM (CBCS)

Applicable to the batch admitted from AY 2024-25 and onwards.

Contents

Regulations for Master of Computer Applications (MCA)
Under Nitte (Deemed to be University)

PREAMBLE

The Department was established in the year 1993 and currently offers two-year MCA program with the intake of 180. The program is recognized by the AICTE and offered at NMAM Institute of Technology, Nitte, Off Campus Center of Nitte (Deemed to be University). MCA is an educational program in Computer Applications leading to the award of Degree. It involves events/ activities, comprising of lecturers/ tutorials/ laboratory work/ field work, outreach activities/ project work/ vocational training/ viva/ seminars /internship/ assignments/ presentation/ quiz/ self-study, etc., or a combination of some of these. The core subjects of this program are designed to introduce students to the various areas of Computer Science and Engineering. The departmental laboratories provide good work ambiance with state-of-the-art computing facilities to enable the students to carry out projects and acquire expertise in emerging technologies. The students are offered special inputs in aptitude, soft skills, and overall personality development regularly to enhance their employability.

INTRODUCTION

Regulations for the MCA program under Nitte (Deemed to be University) govern the policies and procedures including selection, admission, imparting of instructions, conduct of examinations, evaluation and certification of candidate's performance and all amendments thereto, leading to the award of MCA degree. The regulations shall come into effect from the academic year 2023-24 and are applicable to the batch admitted from 2023-24 and onwards. This set of regulations shall be binding on all the candidates undergoing the said program.

1. KEY INFORMATION

Program Title	Master of Computer Applications
Short Description	Two-year, Four Semester Choice Based Credit System (CBCS) type of post graduate program as per NEP 2020, with English as medium of Instruction
Program Code	22CMPA20D2
Revision version	2022.05 These regulations may be modified from time to time as mandated by the policies of the University. Revisions are to be recommended by the Board of Studies for Computer Applications and approved by the Academic Council.
Effective from	01-08-2024
Approvals	Approved in the 57th Academic Council meeting of NITTE (Deemed to be University), held on 26.06.2024 and vide Notification Ref: N(DU)/REG/AC-NMAMIT/2023-24/1422 dated 09.07.2024.
Program offered at	N M A M Institute of Technology, Nitte NITTE (Deemed to be University) (OFF CAMPUS CENTRE)
Grievance and dispute resolution	All disputes arising from this set of regulations shall be addressed to the Board of Computer Applications. The decision of the Board of Computer Applications is final and binding on all parties concerned. Further, any legal disputes arising out of this set of regulations shall be limited to jurisdiction of Courts of Mangalore only.

2. ELIGIBILITY FOR ADMISSION

2.1 Entry requirements

Admission options	Entry requirements
Admission into two-year MCA Program	Passed BCA/ bachelor's degree in computer science engineering or equivalent Degree. OR Passed B.Sc./ B.Com./ B.A. with Mathematics at 10+2 level or at Graduation Level (with additional bridge courses as per the norms of the concerned University). Obtained at least 50% marks (45% marks in case of candidates belonging to reserved category) in the qualifying Examination.

* Eligibility will be determined by the University based on evaluation of equivalency of qualification

* The candidates should have appeared for NUCAT Entrance Examination conducted by Nitte (Deemed to be University) [Nitte (DU)].

2.2 Qualifications from foreign countries

Candidates with qualifications from educational institutions outside of India, may be admitted to the program(s) subject to establishment of equivalence by the university. The Program Committee will evaluate and establish the eligibility of such candidates.

3. COURSE

A “Course” is defined as a unit of learning that typically lasts one semester, led by one or more teachers, for a fixed roster of students. Often referred to as a “subject” or “paper”, a course has identified course outcomes, modules / units of study, specified teaching-learning methods and assessment schemes. A course may be designed to include lectures, tutorials, practical, laboratory work, field work, project work, internship experiences, seminars, self-study components, online learning modules etc. in any combination of some of these.

3.1 Types of Courses

The following types of courses are included in the MCA Program:

3.1.1 **Professional Core Courses (PCC):** These are the professional Core Courses, relevant to the chosen specialization/ branch. The core courses shall be compulsorily studied by students, and it is mandatory to complete them to fulfill the requirements of a Program.

3.1.2 **Professional Elective Courses (PEC):** These are Professional Electives, relevant to the chosen specialization/branch and can be chosen from the pool of courses. It shall be supportive to the discipline providing extended scope/enabling exposure to some to other discipline /domain and nurturing student proficiency skills.

3.1.3 **Seminar (SEM):** Each student has to present the seminar on a specific topic chosen from the relevant field /list provided by the department under the supervision of a faculty coordinator.

3.1.4 **Internship (INT):** The internship (a form of experimental learning) program is a workplace based professional learning experience that offers supervised exposure to real life work experience in an area related to field of study or career interest. An internship may be undertaken at a workplace such as industry/ R&D organization / Government organization, or any other reputed organization / institution recognized for the purpose by the University. The internship program not only helps fresh pass-outs in gaining professional know-how but also benefits corporate sectors. The internship also enhances the employability skills of the student passing out from Technical Institutions.

3.1.5 **Project Work (PROJ):** Provide experiential learning opportunities for students. Students are required individually, or in a small group, select and complete a project that may include review, design, development, curation, analysis, etc., with application of skills and knowledge relevant to area of study. Mini-project and Project work carried out at the parent institution, or any university / Government recognized organization/ Industry without affecting the regular class work are such courses.

3.1.6 **Research Methodology (RM):** Helps to understand the meaning, objectives and characteristics of research. Undergoing the course enables a student to select and define research problems. It helps students understand research designs, methods and also ethics in research and publications. The course also provides information on Intellectual property rights.

3.1.7 **Mandatory Non-Credit Course (MNC):** This course is mandatory for the students who have completed their bachelor's degree program in non-computer disciplines.

Typically, MCA program has the following component of courses.

SN	Category	Minimum Credits to be earned for the MCA degree
1	Core Courses (PCC)	57 (43 +14 Lab)
2	Elective Courses (PEC)	15
3	Research Methodology (RM)	02
4	Seminar (SEM)	01
5	Mini Project (PROJ)	04
6	Internship (INT)	03
7	Major Project (PROJ)	18
	TOTAL	100

4. ACADEMIC YEAR

Refers to the sessions of two consecutive semesters (odd followed by an even) including periods of vacation.

5. COURSE REGISTRATION

The faculty advisors will guide the students in registering for courses. Course Registration refers to formal registration for the Courses of a semester by every student under the supervision of a faculty advisor (also called Mentor, Counselor etc.,) in each semester for the Institution to maintain proper record.

6. CREDITS

Refers to a unit by which the Course work is measured. It determines the number of hours of instructions required per week. One credit is equivalent to one hour of lecture or two hours of laboratory/ practical courses/ tutorials/ field work per week etc.

Credit Limits	Regular Semester
Minimum credits per Semester	16
Maximum credits per Semester	28

7. REGISTRATION FOR ELECTIVES

7.1 Elective options will be offered based on feasibility.

7.2 Registration to elective options will be on first-come, first-served basis.

7.3 Registration will be subject to minimum and maximum enrolment requirements specified by the department. Typically, the minimum enrolment requirement is 15 and the maximum enrolment limit is 60.

7.4 If a course option is not offered due to that lack of minimum enrolments, students who pre-register for it will be transferred to other course options, based on their preference, subject to availability.

8. REQUIREMENTS FOR PROGRESSION

8.1 A student may progress from Semester 1 to 2, from Semester 3 to 4 irrespective of the grades obtained in the courses of these odd Semesters.

8.2 A student may carry over not more than 4 courses from 1st& 2nd Semester (put together) to progress to 3rd semester.

9. PROGRAM STRUCTURE

I Semester MCA

SN	Code	Course Type	Subject	Contact Hour/week				Marks		SEE (Hours)	Credits
				L	T	P	SL	CIE	SEE		
1	22MCA101	PCC	Data Structures with Algorithms	3	1	0	2	50	50	3	4
2	22MCA102	PCC	Advanced Database Systems	3	1	0	2	50	50	3	4
3	22MCA103	PCC	Computer Organization and Architecture	3	1	0	2	50	50	3	4
4	22MCA104	PCC	Mathematical Foundation for Computer Applications	3	1	0	2	50	50	3	4
5	22MCA105	PCC	Software Engineering and Testing	3	1	0	1	50	50	3	4
6	22MCA106	RM	Research Methodology and Publication Ethics	2	0	0	1	50	50	3	2
7	22MCA107	PCC	Data Structures with Algorithms Lab	0	0	4	2	50	50	3	2
8	22MCA108	PCC	Advanced Database Systems Lab	0	0	4	2	50	50	3	2
9	22MCA109	MNC	Fundamentals of Programming *	00	0	4	1	50	--	--	0
				(For 2 Weeks)							
			Total	17	5	12		450	400	24	26
	Total Additional Academic activities hours/week			-	-	-	15				
	Learning hours per week			17	5	12	15				
	Total notional learning hours (L+T+P+SL)		Per Week	49							
			Per Semester	49*16 = 784							

*The course 22MCA109, Fundamentals of Programming is a mandatory noncredit Bridge Course only for the students who have completed their Bachelor's degree program in non-computer disciplines.

SL: Supervised Learning

II Semester MCA

SN	Code	Course Type	Subject	Contact Hours/week				Marks		SEE (Hours)	Credits
				L	T	P	SL	CIE	SEE		
1	22MCA201	PCC	Data Communication and Networks	3	0	0	2	50	50	3	3
2	22MCA202	PCC	Enterprise Java	3	0	0	2	50	50	3	3
3	22MCA203	PCC	Operating Systems with UNIX	3	0	0	2	50	50	3	3
4	22MCA204	PCC	Data Warehousing and Data Mining	3	1	0	2	50	50	3	4
5	22MCA205	PCC	Professional Communication Skills	1	1	0	1	50	50	3	2
6	22MCA206	PCC	Data Communication and Networks Lab	0	0	4	2	50	50	3	2
7	22MCA207	PCC	Enterprise Java Lab	0	0	4	2	50	50	3	2
7	22MCA208	PCC	Operating Systems with UNIX Lab	0	0	4	2	50	50	3	2
8	22MCA209	SEM	Technical Seminar and Report Writing	0	0	2	*8	50	--	--	1
9	22MCA21X	PEC	Elective - I	3	0	0	1	50	50	3	3
10	22MCA22X	PEC	Elective - II	3	0	0	1	50	50	3	3
			Total	19	2	14	17	550	500	30	28
	Total Additional Academic activities hours/week			-	-	-	17				
	Learning hours per week			19	2	14	17				
	Total notional learning hours (L+T+P+SL)		Per Week	52							
			Per Semester	52*16 +*8 = 832 +*8 = 840							

Audit Courses			
Sl No	Course Code	Course Name	Duration
1	22MCAAU21	Data Science BootCamp - From Analysing Data to Creating ML Models from Geek For Geeks	25 hours
2	22MCAAU22	Introduction to Japanese Language	25 hours

SL: Supervised Learning

III Semester MCA											
SN	**Code**	**Course Type**	**Subject**	**Contact Hours/week**				**Marks**		**SEE (Hours)**	**Credits**
				L	T	P	**SL**	**CIE**	**SEE**		
1	22MCA301	PCC	Artificial Intelligence and Machine Learning	3	1	0	2	50	50	3	4
2	22MCA302	PCC	Advanced Web Technologies	3	1	0	2	50	50	3	4
3	22MCA303	PCC	Artificial Intelligence and Machine Learning Lab	0	0	4	1	50	50	3	2
4	22MCA304	PCC	Advanced Web Technologies Lab	0	0	4	1	50	50	3	2
5	22MCA305	PROJ	Mini Project	0	1	9	3	50	50	3	4
6	22MCA33X	PEC	Elective – III	3	0	0	1	50	50	3	3
7	22MCA34X	PEC	Elective – IV	3	0	0	1	50	50	3	3
8	22MCA35X	PEC	Elective – V	3	0	0	1	50	50	3	3
			Total	15	3	17	12	400	400	24	25
	Total Additional Academic activities hours/week			-	-	-	12				
	Learning hours per week			15	3	17	12				
	Total notional learning hours (L+T+P+SL)		Per Week	47							
			Per Semester	47*16 = 752							

SL: Supervised Learning

IV Semester MCA									
SN	**Code**	**Course Type**	**Subject**	**Contact Hours/week**	**SL**	**CIE Marks**	**SEE Marks**		**Credits**
							Evaluation	**Viva Voce**	
1	22MCA401	INT	Internship*	Full Time (4 weeks) *160		50	--	50	3
2	22MCA402	PROJ	Major Project**	Full Time (18 weeks) **540		100	100	100	18
			Total	700		150	100	150	21

* The internship should be completed after third semester examinations and before the commencement of fourth semester. Total of 160 hours.
**The Major Project should be carried out preferably in the industry for a minimum duration of 18 weeks with total 540 Learning hours.

Electives:

Elective Group – I	
Code	**Subject**
22MCA211	Digital Image Processing & Pattern Recognition
22MCA212	Environmental Studies and Green IT
22MCA213	Soft Computing
22MCA214	Parallel Processing
22MCA216	Distributed Computing

Elective Group – II	
Code	**Subject**
22MCA221	E-Commerce
22MCA222	Health Care Analytics
22MCA223	Accountancy and Financial Management
22MCA224	Bioinformatics
22MCA226	.NET Framework and C#

Elective Group – III	
Code	**Subject**
22MCA331	Mobile Computing & Application Development
22MCA332	Digital and Social Media Marketing
22MCA335	Software Risk Identification and Management
22MCA336	Industrial and Medical IOT

Elective Group – IV	
Code	**Subject**
22MCA343	Block Chain Technology
~~22MCA344~~	~~Network and Cyber Security~~ (will not be offered or it will be dropped)
22MCA345	Cyber Forensics
22MCA346	Quantum Information and Cryptography
	NPTEL Course (noc25-cs16 - Cryptography and Network Security)

Elective Group – V	
Code	**Subject**
22MCA351	Cloud Computing & Big Data Analytics
22MCA352	Natural Language Processing
22MCA355	Management Information Systems
22MCA356	Time Series Analysis and Prediction

10. ATTENDANCE

10.1 MCA is a full-time program.

10.2 Students are not permitted to enroll in any other program offered by this or other University without prior permission.

10.3 All students must attend every lecture, tutorial and practical classes. To account for approved leave of absence (e.g. representing the University in sports, games or athletics, placement activities, NCC/NSS activities etc.) and/ or any other such contingencies like medical emergencies etc., the attendance requirement shall be a minimum of 85% of the classes conducted.

10.4 A candidate having shortage of attendance in one or more subjects shall have to repeat those courses. Such students shall re-register for the same subjects in the subsequent semester/ academic year. Semesters are subjected to maximum permissible credits.

10.5 A candidate, who does not satisfy the attendance requirement mentioned as above, in minimum number of credit requirement shall not be eligible to appear for the examination of that semester and not promoted to higher semester. The candidate shall be required to repeat that semester along with regular students later.

10.6 If a candidate, for any reason, discontinues the course in the middle, he/she may be permitted to register to continue the course along with subsequent batch, subject to the condition that he/she shall complete the class work, lab work and seminar including the submission of dissertation within maximum stipulated period. Such candidate is not eligible to be considered for the award of rank.

10.7 The Head of the Department shall notify regularly, the list of such candidates who fall short of attendance. The list of the candidates falling short of attendance shall be sent to the Principal with a copy to Controller of Examinations.

11. ABSENCE DURING THE SEMESTER

11.1 **Leave of Absence:** If the period of leave is more than two days and less than three weeks, prior application for leave shall have to be submitted to the Head of the Department concerned, with the recommendation of the Faculty-Advisor stating fully the reasons for the leave request along with supporting documents. It will be the responsibility of the student to intimate the course instructors, Head of the Department and also Chief Warden of the hostel, regarding his/her absence before availing leave.

11.2 **Absence during Mid-Semester Examinations:** A student who has been absent from a Mid-Semester Examination (MSE) due to illness and other contingencies may give a request for additional MSE within two working days of such absence to the office of the respective Head of the Department (HOD) with necessary supporting documents and certification from authorized personnel. The HOD may consider such requests depending on the merits of the case and may permit an additional Mid-Semester Examination for the concerned student.

11.3 Absence during Semester End Examination: In case of absence for a Semester End Examination, on medical grounds or other special circumstances the student can apply for 'I' grade in that course with necessary supporting documents and certifications by authorized personnel to the Controller of Examination through Chairman of The Department. The Controller of Examination may consider the request depending on the merits of the case and permit the make-up Semester End Examination for the concerned student. The student may subsequently complete all course requirements within the date stipulated by DPGC (which may be extended till first week of next semester under special circumstances) and 'I' grade will then be converted to an appropriate letter grade. If such an application for the 'I' grade is not made by the student, then a letter grade will be awarded based on his in-semester performance.

12. WITHDRAWAL FROM THE PROGRAM

12.1 Temporary Withdrawal

A student who has been admitted to a Post Graduate Degree program of the College may be permitted to withdraw temporarily, for a period of one semester or more on the grounds of prolonged illness or grave calamity in the family etc. The student should abide by the applicable rules and regulations of the college/University at the time of Temporary Withdrawal.

12.2 Permanent Withdrawal

Any student who withdraws admission before the closing date of admission for the Academic Session is eligible for the refund of the deposits only. Fees once paid will not be refunded on any account. Once the admission for the year is closed, the following conditions govern withdrawal of admissions: a) A student who wants to leave the College for good, will be permitted to do so (and can take Transfer Certificate from the College, if needed), only after remitting the Tuition fees as applicable for all the remaining semesters and clearing all other dues, if any. b) Those students who have received any scholarship, stipend or other forms of assistance from the College shall repay all such amounts in addition to those mentioned in (a) above. The decision of the Principal of the Institute regarding withdrawal of a student is final and binding.

13. EVALUATION

13.1 Course Evaluation

For all courses evaluation will be based on both formative assessment (Continuous Internal Evaluation, CIE) and summative assessment (Semester End Evaluation, SEE). CIE and SEE will carry 50 % and 50% respectively, to enable each course to be evaluated for 100 marks, irrespective of its credits. Weightage for CIE and SEE will be 50% each.

13.2 Continuous Internal Evaluation (CIE)

CIE refers to the evaluation of students' achievement in the learning process. The course instructor will perform the Continuous Internal Evaluation (CIE) for 50 marks which includes tests, homework, problem solving, group discussion, quiz, mini-project and seminar throughout the semester, with weightage for the different components being fixed at the University level as follows.

1. Quizzes, tutorials, assignments, etc., - 20 marks
2. Mid Semester Examination - 30 marks

An additional MSE may be conducted for those students absent for valid reasons/ with prior permission. For those students who could not score minimum required CIE marks (25 marks), an additional MSE may be conducted, and however the maximum CIE marks shall be restricted to 25 out of 50.

13.3 Semester End Examinations (SEE)

Refers to examination conducted at the University level covering the entire Course Syllabus. The Syllabi is modularized and SEE questions are selected from each module, with a choice confined to the concerned module only. SEE is also termed as university examination.

13.4 For Technical Seminar and Fundamentals of Programming, the evaluation will be based on formative assessment (Continuous Internal Evaluation) only.

13.5 For the major project, the dissertation will be evaluated by two examiners, one of the examiners shall be the guide of the candidate and the other examiner shall be preferably an external expert in the area of the dissertation being evaluated. The evaluation of the dissertation shall be made independently by each examiner.

a) Examiners shall evaluate the dissertation normally within a period of not more than two weeks from the date of receipt of dissertation through email.
b) The examiners shall independently submit the marks for the dissertation during the viva-voce examination date.
c) Sum of marks awarded by the two examiners shall be the final evaluation marks for the dissertation.
d) Viva-Voce examination of the candidate shall be conducted, if the dissertation work and the reports are accepted by the external examiner.
e) If the external examiner finds that the dissertation work is not up to the expected standard and the minimum passing marks cannot be awarded, the dissertation shall not be accepted for SEE.
f) If the dissertation is rejected during the Project work, then the second examiner (external) will be appointed by the COE against whom the candidate has to re-present the same dissertation. The decision of the Second Examiner (external) will be final.
g) If the second examiner (external) accepts the dissertation, then the viva-voce examination of the candidate shall be conducted as per the norms. If the second examiner (external) rejects the dissertation, then the student has to take an extension for a minimum period of 3 months and re-work on the project. After the completion of the extension period, viva-voce examination

of the candidate shall be conducted as per the norms, if the dissertation work is accepted by the external examiner.

h) Viva-voce examination of the candidate shall be conducted jointly by the external examiner and internal examiner/guide at a mutually convenient date.

i) The relative weightages for the evaluation of dissertation and the performance at the viva-voce shall be as per the scheme of teaching and examination.

j) The marks awarded by both the examiners at the viva-voce examination shall be sent jointly to the office of the Controller of Examination immediately after the examination.

k) The candidates who fail to submit the dissertation work within the stipulated time have to apply for the extension of the Project duration through the Guide and the head of the department to the office of the Controller of Examination. Such candidate is not eligible to be considered for the award of rank.

14. THE TRANSITIONAL GRADES

'I', 'W' and 'X' would be awarded in the following cases. These would be converted into one or the other of the letter grades (O to F) after the student completes the course requirements.

14.1 Grade "I": To a student having attendance ≥85% and CIE ≥ 70%, in a course, but remained absent from SEE for valid & convincing reasons acceptable to the College, like:

i. Illness or accident, which disabled him/her from attending SEE.

ii. A calamity in the family at the time of SEE, which required the student to be away from the College.

iii. However, the committee chaired by the Principal is authorized to relax the requirement of CIE ≥ 70% if the student is hospitalized or advised to have long term rest after discharge from the hospital by the Doctor.

iv. Students who remain absent for Semester End Examinations due to valid reasons and those who are absent due to health reasons are required to submit the necessary documents along with their request to the Controller of Examinations to write Make up Examinations within 2 working days of that examination for which he or she is absent, failing which they will not be given permission.

14.2 Grade "W": To a student having satisfactory attendance at classes but withdrawing from that course before the prescribed date in a semester as per faculty advice.

14.3 Grade "X": To a student having attendance ≥85% and CIE ≥ 70%, in a course but SEE performance could result in a 'F' grade in the course. (No "F" grade awarded in this case, but student's performance record will be maintained separately).

14.4 The Make Up Examination facility would be available to students who may have missed attending the SEE of one or more courses in a semester for valid reasons and given the 'I' grade. Also, students having the 'X' grade shall also be eligible to take advantage of this facility. The makeup examination would be held as per dates notified in the Academic Calendar. However, it should be made possible to hold a make-up examination at any other time in the semester with the permission of the Academic Council of the College. In all these cases, the standard of SEE would be the same as the normal SEE.

- Make Up examination will be conducted for the candidates who has a CIE>=35 marks and may have missed to attend the SEE covering the entire course syllabus. The standard of the Make Up Examination is same as that of the SEE.
- All the 'W' grades awarded to the students would be eligible for conversion to the appropriate letter grades only after the concerned students re-register for these courses in a main/summer semester and fulfill the passing standards for their CIE and (CIE+SEE).
- The suggested passing standards are CIE to have >=50% and CIE+SEE to have a grade better or at least equal to C. For maintaining high standards, the students scoring less than 50% in CIE are advised to withdraw and to re-register for the course when offered next. The letter grade ‘W’ to be entered in the grade card against the subject and not to be taken into account while calculating SGPA & CGPA

14.5 SUMMER / FAST TRACK SEMESTER:

i) The students who have satisfied CIE and attendance requirements for the course/s and obtained an F grade in SEE are permitted to appear directly in ensuing examination/s as backlog paper/s. The students need not re-register for such course/s in the summer / fast track semester. In case the student wishes to improve CIE he/she has to re-register for the summer / regular semester as and when offered next.

ii) The student who obtains required attendance and CIE in the summer semester, but obtains an ‘F’ grade in SEE; is permitted to appear for SEE subsequently as backlog course/s. The student need not repeat the course for Attendance and CIE.

iii) The course/s for which the student does not possess satisfactory attendance and CIE score shall be marked as ‘N’ on the grade sheet. Such students are not permitted to SEE for the courses marked as ‘N’ on the grade sheet. The students have to re-register only for course/s marked as ‘N’ in the summer /subsequent semester whenever that course is offered and obtain the required CIE and attendance. Subsequently, they are eligible to appear for SEE in such course/s.

iv) Courses with Transitional Grades viz “W”, “I” and “X” are also eligible to register in the summer semester in case they wish to improve their score in CIE.

v) All courses may not be offered in the summer semester. It is the discretion of the University to offer the courses based on the availability of resources.

The institutes shall notify timetable for the summer semester well in advance.

vi) Summer semester is optional; it is for the student to make the best use of the opportunity.

vii) A student is permitted to register for maximum of 16 credits in the Summer / fast track semester.

viii) A student has to choose those courses which are offered by the Institution in a given summer semester.

ix) In the summer semester, each course needs to be offered for the required number of lectures/ tutorial/ laboratory hours as prescribed in the syllabus.

14.6 SUPPLEMENTARY EXAMINATION: Refers to the examination conducted to assist slow learners and/or failed students through make up courses for duration of 8 weeks. This comprises of both the CIE & SEE and will be conducted after the completion of First year MCA even semester.

15. QUALIFYING STANDARD

- Sessional (CIE): Score: ≥50% (≥25 marks)
- Terminal (SEE): Score: ≥40% (≥20 marks)

For securing a final Pass:

Total 50 % of the Course maximum marks (100) i.e., sum of the CIE and SEE marks prescribed for the Course is desired.

16. GRADING SYSTEM: ABSOLUTE GRADING

The performance of a candidate in a course shall be evaluated according to a Letter Grading System, based on both CIE and SEE. The letter grades (O, A+ A, B+, B, C and F) indicate the level of academic achievement assessed on a 10-point scale (0 to 10) (See Table below).

Letter grade system and corresponding marks range

Marks Range (%)	Grade Point	Letter Grade	Descriptor
90 & above	10	O	Outstanding
80-89	9	A+	Excellent
70-79	8	A	Very Good
60-69	7	B+	Good
55-59	6	B	Above Average
50-54	5	C	Average

CGPA	Classification
7.00-& above	**First Class with Distinction**
6.00-6.99	**First Class**
5.00-5.99	**Second Class**

Below 50	0	F	Fails
Absent	0	F	Ab

CGPA < 5.00	**Fails**

16.1 A candidate shall be considered to have completed a course successfully and earned the credits assigned, if he/she secures an acceptable letter grade in the range O to C.

16.2 The letter grade 'F' in any course implies failure in that course and no credit is earned.

16.3 A candidate having satisfactory attendance at classes and meeting the passing standard at CIE in a course but remained absent from SEE shall be awarded 'I' grade in that course. No credit is earned in such a case.

16.4 Grade point averages: The overall performance of a candidate will be indicated by Grade Point Average (GPA).

16.5 Semester Grade Point Average (SGPA) is computed as follows.

$$\boldsymbol{SGPA} = \frac{[\sum(\boldsymbol{Course\ Credit}) \times (\boldsymbol{Grade\ Point})]}{[\sum(\boldsymbol{Course\ Credit})]}$$

All courses of the semester for which a letter grade has been awarded, including F, will be included in this calculation.

16.6 Cumulative Grade Point Average (CGPA) is computed as follows.

$$\boldsymbol{CGPA} = \frac{[\sum(\boldsymbol{Course\ Credit}) \times (\boldsymbol{Grade\ Point})]}{[\sum(\boldsymbol{Course\ Credit})]}$$

All courses of the all the semesters for which a letter grade has been awarded, excluding F, will be included in this calculation.

16.7 Grade Card: Based on the secured letter grades, grade points, SGPA and CGPA, a grade card for each semester shall be issued. On specific request on paying prescribed fee, a transcript indicating the performance in all semesters may be issued.

16.8 Conversions of Grades into Percentage and Class Equivalence

Conversion formula for the conversion of CGPA into percentage is given below:
Percentage of marks secured, 'P' = CGPA Earned × 10
Illustration: for A CGPA of 8.18:

'P' = CGPA Earned 8.18 × 10 = 81.8 %

17. AWARD OF CLASS

17.1 The candidate, who has passed all the courses prescribed, shall be declared to have passed the program

17.2 A candidate who secures CGPA≥7.00 and above shall be declared to have passed in 'First Class with Distinction'.

17.3 A candidate who secures CGPA≥6.00 or more but less than 7.00 shall be declared to have passed in 'First Class'.

17.4 A candidate who secures CGPA≥ 5.00 or more but less than 6.00 shall be declared to have passed in 'Second Class'.

17.5 An attempt means the appearance of a candidate in one or more courses either in part or full in a particular re-examination including supplementary semester's examinations.

17.6 A candidate who fails in the main examination and passes one or more subjects/courses or all subjects/courses in the supplementary examination, such candidate's attempts shall be considered as multiple attempts.

17.7 If a candidate submits application for appearing for the regular examination but does not appear for any of the courses/subjects in the regular University examination, he can appear for supplementary examination provided other conditions such as attendance requirement, internal assessment marks, etc. are fulfilled and his appearing in the supplementary examination shall be considered as the first attempt.`

18. MERIT CERTIFICATES AND UNIVERSITY GOLD MEDALS

Merit Certificates and University Gold Medals will be awarded on the basis of overall CGPA.

18.1 Only those candidates who have completed the MCA Program and fulfilled all the requirements in the minimum number of years prescribed (i.e.,2 years for master's degree) and who have passed each semester in the first attempt is eligible for the award of Merit Certificates and /or University Gold Medals.

18.2 Award of University Gold Medals, if any, are governed by the specific selection criteria that may be formulated by the University for such Medals / Awards

18.3 Candidates with W, N, I, X and F grades and passes the courses in the supplementary examinations are not eligible for the award of Gold Medal or Merit Certificate.

19. RULES FOR GRACE MARKS

19.1 Grace marks up to 1% of the maximum total marks of the courses for which he/she is eligible and have registered (non-credit courses excluded) in the examination or 10 marks whichever is less shall be awarded to the failed course(s), (with a restriction of a maximum of 5 marks per course) provided on the award of such grace marks the candidate passes in that course(s).

20. CHALLENGE EVALUATION

If a student is not satisfied with the marks allotted to him/her in the semester end examinations, he/she could apply for challenge evaluation within the prescribed time specified. In such cases the answer papers will be valued by the DPGC committee and marks secured by the students in the challenge evaluation will be final.

21. AWARD OF DEGREE REQUIREMENTS

The Degree requirements of a student for the MCA Degree program are as follows:

21.1 The maximum duration for a student to comply with the Degree requirements is 8 semesters from the date of first registration for his first semester.

21.2 A student shall be declared to have completed the Degree of Master of Computer Applications, provided the student has undergone the stipulated course work as per the regulations and has earned the prescribed credits, as per the scheme of teaching and examination of the program.

21.3 A student shall be declared successful at the end of the program for the award of Degree only on obtaining CGPA≥5.00, with none of the courses remaining with F grade.

21.4 In case the CGPA falls below 5.00, the student shall be permitted to appear again for SEE for required number of courses (other than seminar, practical, internship and project) subject to the provision of University, to make up CGPA≥5.0. The student should reject the SEE results of previous attempt and obtain written permission from the Controller of Examinations to reappear to the subsequent SEE.

22. TERMINATION FROM THE PROGRAM/READMISSION

A student shall be required to leave the college without the award of the degree, under the following circumstances:

i. Failing to complete the degree requirements in double the duration of the program.

ii. Based on the disciplinary actions suggested by the Academic Council/Governing Council

23. GRADUATION REQUIREMENTS AND CONVOCATION

23.1 A student shall be declared to be eligible for the award of the Degree if he has Fulfilled Degree Requirements

23.2 No Dues to the College, Departments, Hostels, Library Central Computer Centre and any other center

23.3 No disciplinary action pending against him.

23.4 The award of the Degree must be recommended by the Academic council and approved by Governing Council of Nitte (DU)

23.5 Convocation: Degree will be awarded in person to the students who have graduated during the preceding academic year. Degrees will be awarded in absentia to such students who are unable to attend the convocation. Students are required to apply for the convocation along with the prescribed fees, after having satisfactorily completed all the degree requirements within the specified date in order to arrange for the award of the degree during convocation.

24. PROGRAM OUTCOMES

By the end of the program the student will be able to acquire:

24.1 Disciplinary Knowledge: Master of Computer Applications is the culmination of in-depth knowledge in several branches of Computer Science, Statistics and Mathematics. This also leads to study the related areas such as Information Technology and other allied subjects.

24.2 Communication Skills: Ability to communicate various statistical and mathematical concepts effectively using examples and their geometrical visualization. The skills and knowledge gained in this program will lead to proficiency in analytical reasoning which can be used for modelling and solving of real-life problems.

24.3 Critical thinking and analytical reasoning: The students undergoing this program acquire the ability of critical thinking and logical reasoning and capability of recognizing and distinguishing the various aspects of real-life problems. They acquire analytical skills involving paying attention to details and ability to construct logical arguments using correct technical language related to statistics and ability to translate them with popular language when needed.

24.4 Problem Solving: The Mathematical knowledge gained by the students through this program develops an ability to analyze the problems, identify and define appropriate computing requirements for its solutions, analyze and interpret the data which will help policy makers to take a proper decision. This program enhances students' overall development and also equips them with mathematical modeling ability, problem solving skills, analyze problems related to computer science and exhibit a sound knowledge on data structures and algorithms.

24.5 Research related skills: Undertake research projects by using research skills- preparation of questionnaire, conducting sample survey, research projects using sample survey, sampling technique.

24.6 Information/digital Literacy: Exhibiting strong skills required to program a computer for various issues and problems of day-to-day applications with thorough knowledge of programming languages of various levels.

24.7 Self – directed learning: The student completing this program will develop the ability of working independently and to make an in-depth study of various notions of Mathematics, Statistics and Computer Science.

24.8 Moral and ethical awareness/reasoning: The student completing this program will develop an ability to identify unethical behavior such as fabrication, falsification or misinterpretation of data and adopting objectives, unbiased and truthful actions in all aspects of life in general.

24.9 **Design and Development of Solutions:** Ability to design and develop algorithmic solutions to real world problems and acquiring knowledge on statistics and optimization problems. Establishing excellent skills in applying various design strategies for solving complex problems.

24.10 **Lifelong learning:** This program provides self-directed learning and lifelong learning skills. This program helps the learner to think independently and develop algorithms and computational skills for solving real word problems.

24.11 **Modern Tool Usage:** Identify, select and use modern scientific and IT tools or techniques for modeling, prediction, data analysis and solving problems in the area of Computer Science and making.

Syllabus of III Semester MCA
(Master of Computer Applications)

ARTIFICIAL INTELLIGENCE AND MACHINE LEARNING			
Course Code	**22MCA301**	**Course Type**	**PCC**
Teaching Hours/Week (L:T:P: S)	**3:1:0:0**	**Credits**	**04**
Total Teaching Hours	**50**	**CIE + SEE Marks**	**50+50**
Prerequisite	22MCA104		

Teaching Department: Master of Computer Applications

Course Objectives:

1.	To understand the advantages and types of Artificial Intelligence
2.	To understand machine learning, the need for machine learning, its types and data preprocessing
3.	To understand Regression Models, Decision Trees, Support Vector Machines, Naive Bayesian Learning and Evaluating Regression model performance.
4.	To understand Clustering Techniques, K-NN, Dimensionality Reduction using Principal Component Analysis and Reinforcement Learning algorithms.
5.	To learn deep learning concepts, Neural network, Gradient descent, Back Propagation, and Convolutional Neural Networks.

UNIT - I

Artificial Intelligence:	**10 Hours**
What is AI; Why Artificial Intelligence; Goals of AI; Advantages and disadvantages of AI; Applications of AI, Types of AI	

UNIT - II

Machine learning:	**10 Hours**
What is ML; Need for ML; Challenges and applications of ML; classification: Supervised, Unsupervised, Reinforcement learning, AI vs Machine learning, Data Preprocessing.	

UNIT - III

Regression:	**12 Hours**
Simple, Linear Regression, Multiple Regression, Polynomial Regression, Logistic Regression, Decision Trees, Support Vector Machines, Naive Bayesian Learning, Evaluating Regression model performance.	

Unit - IV

Clustering:	**08 Hours**
K-Means Clustering, Hierarchical Clustering, K-NN, Principal Component Analysis, Reinforcement Learning	

Unit - V

Deep Learning and Artificial Neural Networks:	**12 Hours**
Introduction to deep learning, The Neuron, How do Neural network learn and work? Gradient descent, Back Propagation, ANN in python, Convolutional Neural Networks.	

Course Outcomes: At the end of the course student will be able to

1.	Outline the fundamental concepts of Artificial Intelligence and its types.
2.	Analyze the need for machine learning and data preprocessing
3.	Apply Regression Models, Decision Trees, Support Vector Machines, Naive Bayesian Learning for data analysis and prediction.

4.	Apply machine learning techniques like Clustering Techniques, K-NN, Dimensionality Reduction using Principal Component Analysis and Reinforcement Learning algorithms for data analysis and classification.
5.	Apply deep learning concepts like Neural network, Gradient descent, Back Propagation, and Convolutional Neural Networks for classification.

Course Outcomes Mapping with Program Outcomes & Program Specific Outcomes

Program Outcomes→ ↓ **Course Outcomes**	1	2	3	4	5	6	7	8	9	10	11	12	**PSO1**	**PSO2**
22MCA301.1	3	3	3	3	3	-	3	-	2	3	3	3	3	3
22MCA301.2	3	3	3	3	3	-	3	-	2	3	3	3	3	3
22MCA301.3	3	3	3	3	3	-	3	-	2	3	3	3	3	3
22MCA301.4	3	3	3	3	3	-	3	-	2	3	3	3	3	3
22MCA301.5	3	3	3	3	3	-	3	-	2	3	3	3	3	3

1: Low 2: Medium 3: High

TEXT BOOKS:

1.	Cosma Rohilla Shalizi, Advanced Data Analysis from an Elementary Point of View, 2015
2.	Tom M Mitchell, “Machine Learning”, 1st Edition, McGraw Hill Education, 2017.
3.	Elaine Rich, Kevin K and S B Nair, “Artificial Intelligence”, 3rd Edition, McGraw Hill Education, 2017
4.	Saroj Kaushik, Artificial Intelligence, Cengage learning, 2014

REFERENCE BOOKS:

1.	Introduction to Machine Learning with Python - A Guide for Data Scientists (Muller Andreas)
2.	Data Science: Concepts and Practice - By Vijay Kotu, Bala Deshpande 2nd edition, Publisher -Morgan Kaufmann, 2018
3.	Allen B. Downey, “Think Python: How to Think Like a Computer Scientist”, 2nd Edition, Updated for Python 3, Shroff/O‘Reilly Publishers, 2016
4.	**Data Science For Dummies,** Lillian Pierson, John Wiley & Sons, 21-Feb-2017

ADVANCED WEB TECHNOLOGIES

Course Code	**22MCA302**	**Course Type**	**PCC**
Teaching Hours/Week (L:T:P: S)	**3-1-0**	**Credits**	**04**
Total Teaching Hours	**50**	**CIE + SEE Marks**	**50+50**
Prerequisite	**22MCA109**		

Teaching Department: Master of Computer Applications

Course Objectives:

1.	To understand and apply the principles of XHTML, CSS, HTML5, and Bootstrap, and develop the skills to create structured web pages, implement effective styling, leverage the features of HTML5 for modern web development, and utilize Bootstrap for responsive and visually appealing designs.
2.	To understand PHP scripting language and its applications in web development.
3.	To understand JavaScript and its programming concepts, and overview of XML key components.
4	To understand JavaScript execution environment and different event handler techniques.
5.	To understand jQuery concept and its effects and provide an introduction to AngularJS.

UNIT - I

XHTML: — **02 Hours**

Standard Structure, Basic Text Markup, Lists, Tables, Forms, Frames.

CSS: — **03 Hours**

Introduction, Levels of Style Sheets, Selector Forms, Property Value Forms, Font Properties, List Properties, Color, Alignment of Text, The Box Model, The <span> and <div> tags.

HTML5: — **03 Hours**

Introduction of html5, iframes, layout, responsive web design(view port),computer code, new semantics elements in HTML5, Migration, canvas, svg, input types, new form elements and attribute, google map, media(audio, video)

BOOTSTRAP: — **04 Hours**

Introduction of Bootstrap, First Web Page with Bootstrap, Bootstrap Grid system, CDN, Tables, Images, Jumbotron and Page Header, Wells, Alerts, Buttons, Badges and Labels, Progress Bars, List Groups, Navigation Bar, Forms, Form Inputs, Media Objects, Carousel Plug-in

UNIT -II

PHP: — **06 Hours**

Overview of PHP and uses of PHP, General Syntactic Characteristics, Primitives, Operations and expressions, Control Statements, Arrays, Functions, Pattern Matching, Form Handling, Files.

Advanced PHP: — **04 Hours**

Cookies, Session tracking, Database Access with PHP and MySQL.

UNIT - III

JavaScript — **06 Hours**

Overview of JavaScript, Object Orientation and JavaScript, Syntactic Characteristics, Screen Output and Keyboard Input, Control Statements, Object Creation and Modification, Arrays, Functions, Constructors, Pattern Matching.

XML	**04 Hours**
Introduction, Syntax, Document Structure, Document Type Definitions, Namespaces, XML Schemas, Displaying Raw XML Documents, Displaying XML Documents with CSS.	

UNIT - IV

Dynamic Documents with JavaScript	**08 Hours**
JavaScript Execution Environment, The Document Object Model, Element Access in JavaScript, Events and Event Handling, Handling Events from the Body Elements, Button Elements, Text Box and Password Elements, The Navigator Object. Introduction to dynamic documents, Positioning Element, Moving Elements, Element Visibility, Changing Colors and Fonts, Dynamic Content, Stacking Elements, Locating the Mouse Cursor, Reacting to a Mouse Click, Slow Movement of Elements, Dragging and Dropping Elements.	

UNIT - V

JQUERY:	**05 Hours**
Introduction, Syntax, selectors, events, JQuery HTML, Effects	

Introduction to Angular JS:	**05 Hours**
What is AngularJS? Angular Directives, AngularJS Expression, Angular Expression, AngularJS Module, AngularJS Controller, AngularJS Filter.	

Course Outcomes: At the end of the course student will be able to

1.	Create well-structured web pages using XHTML, apply CSS styling techniques, utilize HTML5 features for modern web development, and design responsive websites using Bootstrap, demonstrating proficiency in web development with these technologies.
2.	Develop proficiency in PHP programming by demonstrating knowledge of its manipulate arrays, create functions, handle form data, manage files, utilize cookies and session tracking, and interact with MySQL databases.
3.	Develop proficiency in JavaScript programming as well as gain a solid understanding of XML.
4.	Develop proficiency in JavaScript for creating dynamic web documents.
5.	Effectively use jQuery to manipulate HTML elements and enabling to apply jQuery for dynamic web development and explore further possibilities with AngularJS.

Course Outcomes Mapping with Program Outcomes & Program Specific Outcomes

Program Outcomes→	1	2	3	4	5	6	7	8	9	10	11	12	**PSO**	**PSO**
↓ Course Outcomes													1	2
22MCA302.1	3	2	3	-	3	3	3	3	3	3	3	3	3	-
22MCA302.2	3	2	3	-	3	3	3	3	3	3	3	3	3	-
22MCA302.3	2	2	3	-	3	3	3	3	3	3	3	3	3	-
22MCA302.4	3	2	3	-	3	2	2	2	2	2	2	3	3	-
22MCA302.5	3	2	3	-	3	2	2	2	2	2	2	3	3	-

1: Low 2: Medium 3: High

TEXT BOOKS:

1.	Robert W. Sebesta: Programming with World Wide Web, 4th Edition, Pearson Education, 2008.
2.	Snig Bhaumik: Bootstrap Essentials, PACKT publishing, open source
3.	Bear Bibeault: JQuery in Action, Manning Publications.
4	Nicholas C Zakas et al: Professional AJAX, Wrox publications, 2006.

5	Francis Shanahan: Mashups, Wrox, 2007.	
REFERENCE BOOKS:		
1.	M. Deitel, P.J. Deitel, A. B. Goldberg: Internet & World Wide Web, How to Program, 5th Edition, Pearson Education, 2008	
2	Chris Bates: Web Programming Building Internet Applications, 3rd Edition, Wiley India, 2007.	
3	Xue Bai et al: The Web Warrior Guide to Web Programming, Cengage Learning, 2001	
4	Thomas A. Powel: Ajax The Complete reference, McGraw Hill, 2008.	
5	Gottfried Vossen, Stephan Hagemann: Unleashing Web 2.0 From Concepts to Creativity, Elsevier, 2007.	
6	Steven Holzner : Ajax Bible Wiley India , 2007.	
7	Justin Gehtland et al: A Web 2.0 primer Pragmatic Ajax, SPD Publications, 2006.	
8	Eric Van derVlist et al: Professional Web 2.0 Programming, Wiley India, 2007.	
9	Jake Spurlock: Bootstrap- Responsive Web Development, O'Reilly Media, 2014.	

MOBILE COMPUTING AND APPLICATION DEVELOPMENT			
Course Code	**22MCA331**	**Course Type**	**PEC**
Teaching Hours/Week (L: T: P: S)	**3:0:0:0**	**Credits**	**03**
Total Teaching Hours	**40**	**CIE + SEE Marks**	**50+50**
Prerequisite	**22MCA201**		

Teaching Department: Master of Computer Applications

Course Objectives:

1.	Introduce the student with concept of Mobile Computing and its architecture.
2.	Discuss the evolution of Mobile Communication system over different generations.
3.	Describe various components of Android Mobile Apps and use them in app development.
4.	Design applications using various UI controls supported by Android.
5.	Learn to work with data from various sources for Mobile Applications.

UNIT-I

Introduction to Mobile Computing :	**02 Hours**
Introduction, Architecture for Mobile Computing	

Evolution of Wireless Cellular Networks:	**07 Hours**
GSM: GSM Architecture, Entities, Call routing in GSM, GSM Addresses and Identities, GPRS: Introduction, GPRS Network Architecture, GPRS Network Operations, Third Generation Networks and features, Fourth Generation Networks and features, Fifth Generation technology and benefits.	

UNIT-II

Basics of Android :	**08 Hours**
Introduction to Android, Build your first app, Layouts, Views and Resources, Text and Scrolling Views	

UNIT-III

Activities and Intents:	**08 Hours**
Activities: Activity Life cycle and states, Intents: Implicit intents and Explicit intents	

UNIT-IV

User Experience:	**08 Hours**
User Interaction: input controls, Menus, Screen navigation, Recycler View; Delightful user experience: drawables, styles and themes.	

UNIT-IV

Working in the background:	**03 Hours**
Broadcasts, Services, Notifications	

Working with data :	**04 Hours**
Storing data, Working with SQLite database, Content Providers	

Course Outcomes: At the end of the course student will be able to

1.	Describe the architecture of Mobile Computing and evolution of Mobile Communication technologies.
2.	Develop basic Android applications on Mobile device and emulator.
3.	Summarize Activity Life cycle and develop apps with intents.

4.	Design Android applications with different Android user interface elements to give a delightful experience to the user.
5.	Develop Android applications incorporating databases.

Course Outcomes Mapping with Program Outcomes

Program Outcomes→	1	2	3	4	5	6	7	8	9	10	11	12	**PSO**	**PSO**
↓ Course Outcomes													1	2
22MCA331.1	3	3	2	2	1	-	-	-	-	-	-	1	1	1
22MCA331.2	3	2	-	1	1	-	-	-	-	-	-	-	1	2
22MCA331.3	2	3	3	-	2	-	-	-	-	-	-	-	1	2
22MCA331.4	2	2	2	3	3	-	-	-	-	-	-	-	1	2
22MCA331.5	3	2	3	1	2	-	-	-	-	-	-	-	1	3

1: Low 2: Medium 3: High

TEXTBOOKS:

1.	Dr. Ashok Talukder, Ms. Roopa Yavagal, Mr. Hasan Ahmed: Mobile Computing Technology, Applications and Service Creation, 2nd Edition, Tata McGraw Hill, 2010
2.	Alexander Kukushkin-Introduction to Mobile Network Engineering, 1st Edition, Wiley 2018
3.	Erik Hellman, “Android Programming - Pushing the Limits”, 1st Edition, Wiley India Pvt. Ltd, 2014

REFERENCE BOOKS:

1.	Dawn Griffiths and David Griffiths, “Head First Android Development”, 1st Edition, O’Reilly SPD Publishers, 2015
2.	J F DiMarzio, “Beginning Android Programming with Android Studio”, 4th Edition, Wiley India Pvt Ltd, 2016. ISBN-13: 978-8126565580
3.	Anubhav Pradhan, Anil V Deshpande, “Composing Mobile Apps” using Android, Wiley 2014, ISBN: 978-81-265-4660-2

E Books / MOOCs/ NPTEL

1.	Google Developer Training Material https://google-developer-training.github.io/android-developer-fundamentals-course-concepts-v2/

DIGITAL AND SOCIAL MEDIA MARKETING			
Course Code	**22MCA332**	**Course Type**	**PEC**
Teaching Hours/Week (L: T: P: S)	**3:0:0:0**	**Credits**	**03**
Total Teaching Hours	**40**	**CIE + SEE Marks**	**50+50**

Teaching Department: Master of Computer Applications

Course Objectives:

1.	To learn about basics of digital marketing
2.	To learn concepts on Ad campaigns and internet marketing
3.	To understand the significance of Social media marketing
4.	To learn about SEO, SEM, Web Analytics, Mobile Marketing, Trends in Digital Advertising
5.	To understand the social media strategies and channels.

UNIT - I

Introduction to Digital Marketing :	**08 Hours**

Evolution of Digital Marketing from traditional to modern era, Role of Internet; Current trends, Info-graphics, implications for business & society; Emergence of digital marketing as a tool; Drivers of the new marketing environment; Digital marketing strategy; P.O.E.M. framework, Digital landscape, Digital marketing plan, Digital marketing models.

UNIT - II

Internet Marketing and Digital Marketing :	**08 Hours**

Mix – Internet Marketing, opportunities and challenges; Digital marketing framework; Digital Marketing mix, Impact of digital channels on IMC; Search Engine Advertising: - Pay for Search Advertisements, Ad Placement, Ad Ranks, Creating Ad Campaigns, Campaign Report Generation Display marketing: - Types of Display Ads - Buying Models - Programmable Digital Marketing - Analytical Tools - YouTube marketing.

UNIT - III

Social Media Marketing :	**08 Hours**

Role of Influencer Marketing, Tools & Plan– Introduction to social media platforms, penetration & characteristics; Building a successful social media marketing strategy Facebook Marketing: - Business through Facebook Marketing, Creating Advertising Campaigns, Adverts, Facebook Marketing Tools Linkedin Marketing: - Introduction and Importance of Linkedin Marketing, Framing Linkedin Strategy, Lead Generation through Linkedin, Content Strategy, Analytics and Targeting Twitter Marketing : Introduction to Twitter Marketing, how twitter Marketing is different than other forms of digital marketing, framing content strategy, Twitter Advertising Campaigns Instagram and Snapchat: - Digital Marketing Strategies through Instagram and Snapchat Mobile Marketing: Mobile Advertising, Forms of Mobile Marketing, Features, Mobile Campaign Development, Mobile Advertising Analytics Introduction to social mediametrics

UNIT - IV

Introduction to SEO, SEM, Web Analytics, Mobile Marketing, Trends in Digital Advertising :	**08 Hours**

Introduction and need for SEO, How to use internet & search engines; search engine and its working pattern, On-page and off-page optimization, SEO Tactics - Introduction to SEM Web Analytics: - Google Analytics & Google AdWords; data collection for web analytics, multichannel attribution, Universal analytics, Tracking code Trends in digital advertising

UNIT - V

Social Media Channels:	**08 Hours**

Introduction, Key terms and concepts, Traditional media vs Social media. Social media channels: Social networking. Content creation, Bookmarking & aggregating and Location & social media. Tracking social media campaigns. Social media marketing: Rules of engagement. Advantages and challenges.

Social Media Strategy: Introduction, Key terms and concepts. Using social media to solve business challenges. Step-by-step guide to creating a social media strategy. Documents and processes. Dealing with opportunities and threats. Step-by-step guide for recovering from an online brand attack. Social media risks and challenges

Course Outcomes: At the end of the course student will be able to

1.	Recognize appropriate e-marketing objectives
2.	Appreciate the e-commerce framework and technology.
3.	Illustrate the use of search engine marketing, online advertising and marketing strategies.
4.	Use social media & create templates.
5.	Develop social media strategies to solve business problems.

Course Outcomes Mapping with Program Outcomes & Program Specific Outcomes

Program Outcomes→	1	2	3	4	5	6	7	8	9	10	11	12	**PSO**	**PSO**
↓Course Outcomes													1	2
22MCA332.1	3	2	3	-	-	2	-	2	-	-	2	3	3	2
22MCA332.2	2	3	-	3	-	-	-		-	3	2	-	2	1
22MCA332.3	-	2	-	-	2	-	-	3	-	-	2	2	3	2
22MCA332.4	2	-	-	3	-	2	-	-	2	2	-	-	3	2
22MCA332.5	-	3	-	-	-	2	-	3	-	2	2	-	2	2

1: Low 2: Medium 3: High

TEXT BOOKS:

1.	Seema Gupta "Digital Marketing" Mc-Graw Hill, 1st Edition – 2017
2.	Internet Marketing: Integrating Online and Offline Strategies. M. L. Roberts and Debra Zahay, 3rd Edition, Cengage Publishing, 2013

REFERENCE BOOKS:

1.	Digital Marketing: Strategy, Implementation and Practice, Chaffey D., Ellis-Chadwick, 5th Edition, F., Pearson, 2012
2.	Puneet Singh Bhatia "Fundamentals of Digital Marketing" Pearson Education, 1st Edition–2017

E Books / MOOCs/ NPTEL

1.	https://www.udemy.com/topic/social-media-marketing/

SOFTWARE RISK IDENTIFICATION AND MANAGEMENT

Course Code	**22MCA335**	**Course Type**	**PEC**
Teaching Hours/Week (L: T: P: S)	**3:0:0:0**	**Credits**	**03**
Total Teaching Hours	**40**	**CIE + SEE Marks**	**50+50**
Prerequisite	**22MCA105**		

Teaching Department: Master of Computer Applications

Course Objectives:

1.	Understand the concepts and principles of Risk Management
2.	Apply the concepts and principles of Continuous Risk Management
3.	To learn about the activity planning and risk management principles.
4.	To develop skills to manage the various supporting practices.
5.	To Deploy Risk management and various environmental classes

UNIT - I

08 Hours

Introduction, risk items, risk resolution techniques, and heuristics risk list risk-action list, risk-strategy model, risk-strategy, analysis.

UNIT - II

08 Hours

A Holistic Vision of Software Risk Management, Temporal Dimension, Methodological Dimension, Human Dimension, Graphic Representation of the Holistic Vision of Software Risk Management

UNIT - III

08 Hours

Software Risk Management Methodologies, Basic constructs to Risk Management, Risk Management Paradigm, Risk Taxonomy, Risk Clinic

UNIT - IV

08 Hours

Supporting practices, Software Risk Evaluation (SRE) practice, Continuous Risk Management (CRM), Team Risk Management (TRM), Methodological framework for Software Risk Management (SRM), Software Capability Maturity Model (SW-CMMSM), Software Acquisition-Capability Maturity Model (SA-CMMSM)

UNIT - V

08 Hours

Deployment of the SEI Risk Management program, Major classes within the Hierarchy, Major elements of Risk within each class, Major attributes within each element and class, Product engineering class, Development environment class, Program constraints class

Course Outcomes: At the end of the course student will be able to

1.	Explain Risk Management principles
2.	Estimate the risks involved in various project activities
3.	Gain extensive knowledge about activity planning in Risk management
4.	Use key methods and tools
5.	Adapt Continuous Risk Management to a project.

Course Outcomes Mapping with Program Outcomes & Program Specific Outcomes

Program Outcomes→ ↓Course Outcomes	1	2	3	4	5	6	7	8	9	10	11	12	PSO 1	PSO 2
22MCA335.1	3	2	3	-	-	2	-	2	-	-	2	3	3	2
22MCA335.2	2	3	-	3	-	-	3	-	-	3	2	-	2	1
22MCA335.3	-	2	-	-	2	-	-	3	-	-	2	2	3	2
22MCA335.4	2	-	-	3	-	2	-	-	2	2	-	-	3	2
22MCA335.5	-	3	-	-	-	2	-	3	-	2	2	2	2	2

1: Low 2: Medium 3: High

TEXT BOOKS:

1.	Barry W. Boehm, "Software risk: management principles and practices", reprinted from vol-8.
2.	Implementing Enterprise Risk Management, John Wiley & Son, Inc., Hoboken, New Jersey. John R.S. Fraser, Betty J. Simkins, Kristina Narvaev, 2015

REFERENCE BOOKS:

1.	Robert K. Wysocki ―Effective Software Project Management‖ – Wiley Publication, 2011.
2.	Risk management notes [Online]. Available: http://agile.csc.ncsu.edu/SEMaterials/RiskManagement.pdf

E Books / MOOCs/ NPTEL

1.	https://www.udemy.com/course/introduction-to-software-project-risk-management/

INDUSTRIAL AND MEDICAL IOT

Course Code	**22MCA336**	**Course Type**	**PEC**
Teaching Hours/Week (L:T:P:S)	**3:0:0:0**	**Credits**	**03**
Total Teaching Hours	**40**	**CIE + SEE Marks**	**50+50**
Prerequisite	**22MCA201**		

Teaching Department : Master of Computer Applications

Course Objectives:

1.	Introduce evolution of internet technology and need for IoT with its characteristics.
2.	Discuss on IoT reference layer and various protocols and software with its application.
3,	Train the students to build IoT systems using arduino, Raspberry Pi and open source IoT platforms.
4.	Make the students to apply IoT data for business solution in industrial IoT.
5.	Understand the applications of medical IoT in various domain in secured manner.

UNIT - I	
	04 Hours
Introduction to IoT: Defining IoT, Characteristics of IoT, Sensors, types of sensors, actuator and smart object **Physical design of IoT :** Things in IoT, IoT protocol.	
	04 Hours
Logical design of IoT : Functional blocks of IoT, Communication models & APIs. **IoT & M2M :** Machine to Machine, Difference between IoT and M2M	
UNIT - II	
IoT levels & Deployment templates	**04 Hours**
Developing IoTs : IoT design methodology, Case study on IoT system for weather monitoring.	
Applications of IoT :	**04 Hours**
Case study of home automation, Case study of smart parking and Case study of smart irrigation**.**	
UNIT - III	
IoT Physical device & Endpoint :	**09 Hours**
Basic building blocks of an IoT device, Introduction to Raspberry Pi, About board, Programming raspberry Pi with python, Introduction to arduino board, Programming arduino device.	
UNIT - IV	
Industrial IoT :	**07 Hours**
What is IIOT? IOT Vs. IIOT, History of IIOT, Benefits of IIoT IIoT-An analysis framework-review of existing IoT taxonomies, industry sector, location, connectivity. Case Study	
UNIT - V	

Internet of Medical Things :	**08 Hours**
IoMT and Telehealth, Relation of IoMT with Telehealth and Telemedicine, Benefits of IoMT, IoMT Provider Benefits, Challenges of implementing IoMT, IoMT Devices, Importance of security for IoMT, IoMT Vs IoT.	

Course Outcomes: At the end of the course student will be able to

1.	Classify the protocols of Internet of Things and elaborate the characteristics.
2.	Explain the design methodology of IoT with its application.
3,	Interpret the working of Raspberry Pi and Arduino board and design the models.
4.	Discuss the functionalities of industrial IoT.
5.	Characterize the applications of medical IoT.

Course Outcomes Mapping with Program Outcomes & Program Specific Outcomes

Program Outcomes→ **↓ Course Outcomes**	1	2	3	4	5	6	7	8	9	10	11	12	**PSO** 1	**PSO** 2
22MCA336.1	3	3	3	-	-	2	1	-	3	-	-	-	3	3
22MCA336.2	3	3	3	3	3	2	-	-	1	1	2	-	3	3
22MCA336.3	3	3	2	3	3	2	-	-	1		2	-	3	3
22MCA336.4	2	3	2	2	3	-	-	-	-	-	-	-	3	3
22MCA336.5	3	3	2	2	3	-	-	-	-	-	-	-	3	3

1: Low 2: Medium 3: High

TEXT BOOKS:

1.	Vijay Madisetti, Arshdeep Bahga, “Internet of Things: A Hands -On Approach”
2.	David Hanes, Gonzalo Salgueiro, Rob Barton "IoT Fundamentals: Networking Technologies, Protocols and Use Cases for the Internet of Things", 2019

REFERENCE BOOKS:

1.	Francis daCosta, “Rethinking the Internet of Things: A Scalable Approach to Connecting Everything”, 1st Edition, Apress Publications, 2013
2.	Waltenegus Dargie, Christian Poellabauer, "Fundamentals of Wireless Sensor Networks: Theory and Practice"
3.	Hugh Boyes⁎, Bil Hallaq, Joe Cunningham, Tim Watson “The industrial internet of things (IIoT): An analysis framework”. Published by Elsevier

E Books / MOOCs/ NPTEL

1.	https://www.splunk.com/en_us/data-insider/what-is-the-internet-of-medical-things-iomt.html#iomt-and-telehealth

BLOCK CHAIN TECHNOLOGY

Course Code	**22MCA343**	**Course Type**	**PEC**
Teaching Hours/Week (L: T: P:S)	**3:0:0:0**	**Credits**	**03**
Total Teaching Hours	**40**	**CIE + SEE Marks**	**50+50**

Teaching Department: Master of Computer Applications

Course Objectives:

1.	To understand Block Chain- Blockchain Architecture and its Applications.
2.	To understand crypto currencies- Types and Applications
3.	To understand Concept of Double Spending- Hashing- Mining- payment verification-Resolving Conflicts and Creation of Blocks
4.	To understand Crypto currency wallets and conversion to Fiat Currency.
5.	To understand Smart contracts- usage- application- working principle- law and regulations.

UNIT-I

Introduction — **08 Hours**

Introduction to Blockchain, How Blockchain works, Blockchain vs Bitcoin, Practical applications, public and private key basics, pros and cons of Blockchain, Myths about Bitcoin.

UNIT-II

Blockchain Architecture — **08 Hours**

Blockchain Architecture, versions, variants, use cases, Life use cases of blockchain, Blockchain vs shared Database, Introduction to crypto currencies, Types, Applications.

UNIT-III

Double Spending — **08 Hours**

Concept of Double Spending, Hashing, Mining, Proof of work. Introduction to Merkel tree, Privacy, payment verification, Resolving Conflicts, Creation of Blocks.

UNIT - IV

Introduction to Bitcoin — **08 Hours**

Introduction to Bitcoin, key concepts of Bitcoin, Merits and De Merits Fork and Segwits, Sending and Receiving bitcoins, choosing bitcoin wallet, Converting Bitcoins to Fiat Currency.

UNIT-V

Introduction to Ethereum — **08 Hours**

Introduction to Ethereum, Advantages and Disadvantages, Ethereum vs Bitcoin, Introduction to Smart contracts, usage, application, working principle, Law and Regulations. Case Study

Course Outcomes: At the end of the course student will be able to

1.	Analyze and explain Block Chain, blockchain architecture and its applications.
2.	Explore and explain crypto currencies, types and applications.
3.	Examine and explain the concept of double spending, hashing, mining, payment verification, resolving conflicts and creation of blocks.
4.	Explore crypto currency wallets and conversion of crypto currency to fiat currency.
5.	Analyze and explain smart contracts, usage, application, working principle, law and regulations in relation to crypto currency.

Course Outcomes Mapping with Program Outcomes

Program Outcomes→	1	2	3	4	5	6	7	8	9	10	11	12	PSO	PSO

↓ Course Outcomes													1	2
MCA343.1	3	2	-	3	3	-	-	-	3	-	-	3	3	1
MCA343.2	2	3	-	2	2	-	-	-	2	-	-	3	3	1
MCA343.3	3	3	-	3	3	-	-	-	2	-	-	3	3	1
MCA343.4	2	2	-	3	2	-	-	-	3	-	-	3	3	1
MCA343.5	2	3	-	2	3	-	-	-	2	-	-	3	3	1

1: Low 2: Medium 3: High

TEXTBOOKS:

1.	Arshdeep Bikramaditya Signal, Gautam Dhameja, “Beginning Blockchain: A Beginner's Guide to Building Blockchain Solutions”, APress.
2.	Bahga, Vijay Madisetti, “Blockchain Applications: A Hands-On Approach”. 31 January 2017
3.	Melanie Swan, “Blockchain”, Oreilly

REFERENCE BOOKS:

1.	Aravind Narayan. Joseph Bonneau, “Bitcoin and Crypto currency Technologies”, Princeton
2.	Arthu.T Books, “Bitcoin and Blockchain Basics: A non-technical introduction for beginners”

NETWORK AND CYBER SECURITY

Course Code	**22MCA344**	**Course Type**	**PEC**
Teaching Hours/Week (L: T: P:S)	**3:0:0:0**	**Credits**	**03**
Total Teaching Hours	**40**	**CIE + SEE Marks**	**50+50**

Teaching Department: MCA

Course Objectives:

1.	Understand the concepts of network and cyber security.
2.	Apply key principles of cryptography.
3.	Understand the concept of authentication.
4.	Understand basic principles of cybercrime.
5.	Apply tools to achieve security in cybercrime.

UNIT-I

Introduction to network and cyber security : **10 Hours**

What is network security, Attacks on Computers and Computer Security: Need for Security, Security Approaches, Principles of Security Types of Attacks. What is cyber security, Substitution Techniques: Caesar Cipher, Monoalphabetic Cipher, Playfair Cipher, Hill Cipher, Polyalphabetic Cipher, One Time Pad, rotor machines, Steganography

UNIT-II

Public Key Cryptography : **08 Hours**

Prime Numbers, Fermats and Eulers theorem, testing to Primality: Millar-Rabin algorithm, Principles of Public Key Cryptosystem, RSA algorithm, Diffie-Hellman, Key exchange.
Hash Functions : Applications, Requirements and Security, Hash functions based on Cipher Block chaining

UNIT-III

Authentication of Systems : **07 Hours**

Remote user-authentication principles, Kerberos Version 4, Kerberos Version 5: Environmental Shortcomings, X.509 Authentication Service: certificates, X.509 version3.

UNIT-IV

Introduction to Cybercrime : **07 Hours**

Cybercrime: Definition and Origins of the Word Cybercrime and Information Security, Who are Cybercriminals? Classifications of Cybercrimes, Cybercrime: The Legal Perspectives, Cyberstalking, Cybercafe and Cybercrimes, Botnets: The Fuel for Cybercrime, Attack Vector.

UNIT-V

Tools and Methods Used in Cybercrime : **04 Hours**

Introduction , Proxy Servers and Anonymizers, Phishing, Password Cracking, Key loggers and Spywares , Virus and Worms, Trojan Horses and Backdoors, Steganography, DoS and DDoS Attacks, SQL Injection, Buffer Overflow, Attacks on Wireless Networks, Digital Signatures and the Indian IT Act, Cybercrime and Punishment, Cyberlaw, Technology and Students: Indian Scenario

Course Outcomes: At the end of the course student will be able to

1.	Solve problems on cryptographic techniques.
2.	Implement algorithms related to public key cryptography.

3.	Explain basic authentication principles.
4.	Discuss various cyber crimes.
5.	Identify cybercrime threats using tools.

Course Outcomes Mapping with Program Outcomes & PSO

Program Outcomes→ **↓ Course Outcomes**	1	2	3	4	5	6	7	8	9	10	11	12	**PSO** 1	**PSO** 2
22MCA344.1	2	2	2	1	1	-	-	-	-	-	-	-	1	2
22MCA344.2	2	2	2	1	-	-	-	-	-	-	-	-	3	2
22MCA344.3	2	2	2	1	3	-	-	-	-	-	-	-	3	2
22MCA344.4	2	2	2	1	1	-	-	-	-	-	-	-	3	2
22MCA344.5	2	2	2	1	2	-	-	-	-	-	-	-	3	2

1: Low 2: Medium 3: High

TEXTBOOKS:

1.	William Stallings, "Cryptography and Network Security – Principles and Practices", Prentice Hall of India, 6th Edition
2.	Sunit Belapure and Nina Godbole, "Cyber Security: Understanding Cyber Crimes,Computer Forensics And Legal Perspectives", Wiley India Pvt Ltd, ISBN: 978-81-265-21791 Publish Date 2013
3.	Cryptography and Network Security, Atul Kahate, TMH, 2003

REFERENCE BOOKS

1.	Charlie Kaufman, Radia Perlman, Mike Speciner: Network Security-Private Communication in Public World, 2nd Edition, Pearson Education, 2003

E Books / MOOCs/ NPTEL

1.	https://nptel.ac.in/courses/106106129

CYBER FORENSICS

Course Code	**22MCA345**	**Course Type**	**PEC**
Teaching Hours/Week (L: T: P: S)	**3:0:0:0**	**Credits**	**03**
Total Teaching Hours	**40**	**CIE + SEE Marks**	**50+50**
Prerequisite	**22MCA201**		

Teaching Department: Master of Computer Applications

Course Objectives:

1.	To study the fundamentals of Computer Forensics.
2.	To learn to analyze and validate forensics data.
3.	To study the tools and tactics associated with Cyber Forensics.
4.	Understand the principles of web security and to guarantee a secure network by monitoring and analyzing the nature of attacks through cyber/computer forensics software/tools.
5.	To learn technical aspects & legal aspects related to cyber crime.

UNIT - I

Introduction To Computer Forensics : **08 Hours**

Introduction to Traditional Computer Crime, Traditional problems associated with Computer Crime. Introduction to Identity Theft & Identity Fraud. Types of CF techniques - Incident and incident response methodology - Forensic duplication and investigation. Preparation for IR: Creating response tool kit and IR team. - Forensics Technology and Systems - Understanding Computer Investigation – Data Acquisition.

UNIT - II

Evidence Collection And Forensics Tools : **08 Hours**

Processing Crime and Incident Scenes – Working with Windows and DOS Systems. Current Computer Forensics Tools: Software/ Hardware Tools.

UNIT - III

Analysis And Validation : **08 Hours**

Validating Forensics Data – Data Hiding Techniques – Performing Remote Acquisition – Network Forensics – Email Investigations – Cell Phone and Mobile Devices Forensics

UNIT - IV

Ethical Hacking : **08 Hours**

Introduction to Ethical Hacking – Foot printing and Reconnaissance - Scanning Networks - Enumeration – System ETHICAL HACKING IN WEB Hacking - Malware Threats - Sniffing

UNIT - V

Ethical Hacking In Web : **08 Hours**

Social Engineering - Denial of Service - Session Hijacking - Hacking Web servers - Hacking Web Applications – SQL Injection - Hacking Wireless Networks - Hacking Mobile Platforms.

Course Outcomes: At the end of the course student will be able to

1.	Explain the basics of computer forensics
2.	Adapt a number of different computer forensic tools to a given scenario
3.	Validate forensics data
4.	Identify the vulnerabilities in a given network infrastructure
5.	Implement real-world hacking techniques to test system security

Course Outcomes Mapping with Program Outcomes & Program Specific Outcomes

Program Outcomes→ ↓Course Outcomes	1	2	3	4	5	6	7	8	9	10	11	12	PSO 1	PSO 2
22MCA345.1	3	3	-	-	3	-	-	-	-	3	-	3	3	2
22MCA345.2	3	3	-	-	2	-	-	-	-	3	-	3	3	2
22MCA345.3	3	2	-	-	2	-	-	-	-	3	-	3	3	2
22MCA345.4	3	2	-	-	2	-	-	-	-	2	-	2	3	2
22MCA345.5	3	3	-	-	3	-	-	-	-	3	-	3	3	2

1: Low 2: Medium 3: High

TEXT BOOKS:

1. Bill Nelson, Amelia Phillips, Frank Enfinger, Christopher Steuart, — Computer Forensics and Investigations, Cengage Learning, India Edition, 2016

2. CEH official Certified Ethical Hacking Review Guide, Wiley India Edition, 2015

REFERENCE BOOKS:

1. John R.Vacca, - Computer Forensics, Cengage Learning, 2005

2. Marjie T.Britz, - Computer Forensics and Cyber Crime: An Introduction, 3rd Edition, Prentice Hall, 2013

3. Ankit Fadia - Ethical Hacking 2nd Edition, Macmillan India Ltd, 2006

4. Kenneth C.Brancik - Insider Computer Fraud Auerbach Publications Taylor & Francis Group–2008

E Books / MOOCs/ NPTEL

1. Digital Forensics With Open Source Tools: Using Open Source Platform Tools for Performing Computer Forensics on Target Systems: Windows, MAC, Linux

https://www.pdfdrive.com/digital-forensics-with-open-source-tools-using-open-source-platform-tools-for-performing-computer-forensics-on-target-systems-windows-mac-linux-unix-etc-d156711094.html

QUANTUM INFORMATION AND CRYPTOGRAPHY			
Course Code	**22MCA346**	**Course Type**	**PEC**
Teaching Hours/Week (L:T:P: S)	**3:0:0:0**	**Credits**	**03**
Total Teaching Hours	**40**	**CIE + SEE Marks**	**50+50**
Prerequisite	22MCA104		

Teaching Department: Master of Computer Applications

Course Objectives:

1.	To understand basics of Cryptography with encryption techniques.
2.	To be able to secure a message over insecure channel by various means.
3.	To provide an in-depth understanding of cryptography theories, algorithms and systems.
4.	To defend the security attacks on information systems with secure algorithms
5.	To understand the fundamental concepts on quantum computing.

UNIT - I

Foundations of Cryptography :	**04 Hours**

Information Security-Confidentiality, Integrity and Availability-Authentication, Authorization and Non repudiation-Introduction to Plain Text, Cipher Text, Encryption and Decryption Techniques, Secure Key, Hashing, Digital signature.

Classical Encryption techniques :	**04 Hours**

Symmetric cipher model, substitution techniques, transposition techniques.

UNIT - II

Conventional Symmetric Encryption Algorithms :	**04 Hours**

Block cipher principles, Feistel Cipher Network Structures, Data Encryption Standard, Modes of Operation (ECB, CBC, OFB, CFB), Strength of DES.

Modern Symmetric Encryption Algorithms :	**04 Hours**

Blowfish, Key Distribution: Scenario, Hierarchical Key control, Session Key lifetime, Transparent Key control, Decentralized key control.

UNIT - III

Public Key Cryptography :	**04 Hours**

Prime Numbers, Principles of public key Cryptosystems, RSA algorithm.

Other public key Cryptosystems:	**04 Hours**

Diffie-Hellman Key exchange Algorithms, Elgamal Cryptographic system, Elliptic curve cryptography.

UNIT - IV

Message Authentication and Message Digest :	**04 Hours**

Authentication Requirements, Authentication Functions: Message Encryption, Message Authentication Code, Hash Function, MD5: logic, Compression function

Digital Signatures :	**04 Hours**

Requirements, Direct digital signature, Arbitrated digital signature, Digital Signature Standard (DSS and DSA)

UNIT - V

Quantum Information and Computing:	**08 Hours**
What is Quantum computing, Superposition and Entanglement, Quantum Computer, What are Quantum Bits, History of Quantum Computing, Applications of Quantum Computing, Classical Computing Vs. Quantum Computing, Future of Quantum Computing	

Course Outcomes: At the end of the course student will be able to

1.	Provide security of the data over the network.
2.	Do research in the emerging areas of cryptography and network security.
3.	Develop cryptographic algorithms for information security
4.	Protect any network from the threats in the world.
5.	Explain the basic concepts on quantum computing.

Course Outcomes Mapping with Program Outcomes & Program Specific Outcomes

Program Outcomes→	1	2	3	4	5	6	7	8	9	10	11	12	**PSO**	**PSO**
↓ Course Outcomes													1	2
22MCA346.1	3	3	-	3	3	-	-	-	-	-	-	-	3	2
22MCA346.2	3	3	3	3	3	-	-	-	-	-	3	-	3	2
22MCA346.3	3	3	3	3	2	-	-	-	-	-	3	-	3	2
22MCA346.4	3	2	2	3		-	-	-	-	-	2	-	3	2
22MCA346.5	3	3	3	-	-	-	-	-	-	-	-	-	3	2

1: Low 2: Medium 3: High

TEXT BOOKS:

1.	William Stallings, "Cryptography and Network Security – Principles and Practices", Prentice Hall of India, 7th Edition, 2017 (ISBNNo.:978-0-13-44446-11)
2.	D.R. Stinson, Cryptography: Theory and Practice, 3rd Edition, Boca Raton, FL: Chapman & Hall/CRC, 2005. (ISBNNo.:978-1-58-488508-5)

REFERENCE BOOKS:

1.	Charlie Kaufman, Radia Perlman, Mike Speciner : Network Security-Private Communication in Public World, 2nd Edition Pearson Education
2.	Atul Kahate, Cryptography and Network Security, Tata McGraw Hill

E Books / MOOCs/ NPTEL

1.	https://www.javatpoint.com/what-is-quantum-computing

CLOUD COMPUTING AND BIG DATA ANALYTICS

Course Code	**22MCA351**	**Course Type**	**PEC**
Teaching Hours/Week (L:T:P:S)	**3:0:0:0**	**Credits**	**03**
Total Teaching Hours	**40**	**CIE + SEE Marks**	**50+50**
Prerequisite	22MCA102, 22MCA202		

Teaching Department: Master of Computer Applications

Course Objectives:

1.	Understand the types, characteristics and challenges with digital data
2.	Understand the concept of Big data Analytics and technology landscape
3.	Understand the use of open source software framework called Hadoop
4.	Understand different cloud computing service models and cloud security
5.	Learn about multicloud computing and Cloud Computing in Business

UNIT - I

Types of Digital Data	**04 Hours**

Classification of Digital Data, Structured Data, Sources of Structured Data, Ease of Working with Structured Data, Semi-Structured Data, Sources of Semi-Structured Data, Unstructured Data, Issues with "Unstructured" Data, How to Deal with Unstructured Data.

Introduction to Big Data	**04 Hours**

Characteristics of Data, Evolution of Big Data, Definition of Big Data, Volume, Velocity, Variety, Challenges of Big Data, Other Characteristics of Data Which are Not Definitional Traits of Big Data, Traditional Business Intelligence (BI) versus Big Data, A Typical Data Warehouse Environment, A Typical Hadoop Environment, Coexistence of Big Data and Data Warehouse.

Unit - II

Introduction to Big Data Analytics	**08 Hours**

Big Data Analytics, Classification of Analytics, Greatest Challenges that Prevent Businesses from Capitalizing on Big Data, Top Challenges Facing Big Data, Importance of Big Data Analytics, Terminologies Used in Big Data Environment, In Memory Analytics, In Database Processing, Symmetric Multiprocessor System, Massively Parallel Processing, Difference between Parallel versus Distributed Systems, Shared Nothing Architecture, Consistency, Availability, Partition Tolerance (CAP), Basically Available Soft State Eventual Consistency (BASE)

Unit - III

Hadoop Techniques	**07 Hours**

Introducing Hadoop, RDBMS versus Hadoop, Hadoop versus SQL, Distributed Computing Challenges, Key Aspects of Hadoop, Hadoop Components. Features of Hadoop, Key Advantages of Hadoop, Versions of Hadoop, Hadoop 1.0, Hadoop 2.0, Overview of Hadoop Ecosystems, High Level Architecture of Hadoop, Hadoop Distributed File System, HDFS Daemons, Anatomy of File Read, Anatomy of File Write, Processing Data with Hadoop, MapReduce Daemons, MapReduce Example, Managing Resources and Application with Hadoop YARN, Limitations of Hadoop 1.0 Architecture, HDFS Limitation, Hadoop 2: HDFS, Hadoop 2 YARN: Taking Hadoop Beyond Batch, Hadoop Ecosystem

Unit - IV

Cloud Computing definition, Cloud service models (IaaS, PaaS & SaaS) :	**08 Hours**

.

Cloud deployment models (Public, Private, Hybrid and Community Cloud), Private & Public Cloud Definition, Characteristics of Private Cloud, Private Cloud deployment models, Private Cloud Building blocks namely Physical Layer, Virtualization Layer, Cloud Management Layer, When to opt for Public Cloud, Public Cloud Service Models

Unit - V

Cloud Management System	**08 Hours**

Cloud Computing in terms of Application Security, Server Security, and Network Security, multi-cloud management, Management System (e.g. RightScale Cloud Management System), Cloud Computing in Business.

Course Outcomes: At the end of the course student will be able to

1.	Classify the digital data
2.	Recognize the top challenges facing big database
3.	Study components and advantages of Hadoop
4.	Have knowledge of Cloud deployment models
5.	Have knowledge of Server Security, Network Security and Application Security.

Course Outcomes Mapping with Program Outcomes & Program Specific Outcomes

Program Outcomes→	1	2	3	4	5	6	7	8	9	10	11	12	**PSO**	**PSO**
↓ Course Outcomes													1	2
22MCA351.1	2	3	2	3	-	-	3	-	-	-	-	3	1	1
22MCA351.2	3	3	-	-	-	2		-	-	-	-	2	2	2
22MCA351.3	2	-	3	-	1	-	-		1	-	-	2	3	3
22MCA351.4	1	2	-	3		2	1	-	-	-	-	2	2	3
22MCA351.5	2	-	2	3	-	-	-	3	-	-	2	3	3	3

1: Low 2: Medium 3: High

TEXT BOOKS:

1.	Seema Acharya and Subhashini C: Big Data and Analytics, First Edition, Wiley India Pvt. Ltd, 2015
2.	Cloud Computing: Principles and paradigms By Raj Kumar Buyya, James Broberg, Andrezei M. Goscinski, 2011

REFERENCE BOOKS:

1.	Judith Hurwitz, Alan Nugent, Fern Halper, Marcia Kaufman: Big data for dummies
2.	Tom White: Hadoop – The Definitive Guide
3.	Chuck Lam: Hadoop in action
4.	Dirk Deroos, Paul C. Zikopoulos, Roman B. Melnyk, Bruce Brown: Hadoop for dummies

E Books / MOOCs/ NPTEL

1.	Big Data computing-nptel
2.	Cloud Computing and Big Data, C. Catlett, W. Gentzsch, L. Grandinetti, IOS Press, 2013

NATURAL LANGUAGE PROCESSING			
Course Code	**22MCA352**	**Course Type**	**PEC**
Teaching Hours/Week (L:T:P:S)	**3:0:0:0**	**Credits**	**03**
Total Teaching Hours	**40**	**CIE + SEE Marks**	**50+50**

Teaching Department: Master of Computer Applications

Course Objectives:

1.	To learn the basics of Natural Language Processing
2.	Learn the finite-state morphological parsing process
3.	To know about Syntactic Parsing
4.	Understand the basic feature systems for English
5.	To know the concept of Named entity recognition

UNIT - I

Introduction to Natural Language Processing:	**04 Hours**
The study of Language, Applications of NLA, Different levels of language analysis	
Linguistic Background :	**04 Hours**
An outline of English syntax, The elements of simple noun phrases, verb phrases and simple sentences. Morphological Analysis: Regular Expressions, Word Tokenization, Top- Down & Bottom-Up Tokenization, Lemmatization, Text Normalization.	

UNIT - II

Detecting and correcting spelling errors:	**04 Hours**
Minimum edit distance, N-Grams - Counting words in corpora, Simple(un-smoothed) n-grams	
Part of Speech Tagging - English word classes :	**04 Hours**
Tag sets for English, Hidden Markov Models - Markov chains, and The Hidden Markov Model, Viterbi Algorithm.	

UNIT - III

Syntactic Parsing:	**04 Hours**
Grammars and syntax structure, A top down parser	
Depth first strategy vs Breadth first strategy :	**04 Hours**
Bottom up chart parser, Efficiency considerations, Transition Network Grammars, Top down chart parser.	

UNIT - IV

Features and Augmented Grammars :	**04 Hours**
Feature systems and augmented grammars	
Basic feature systems for English :	**04 Hours**
person and number features; Binary features; Morphological analysis and the lexicon; A simple grammar using features	

UNIT - V

Applications:	**04 Hours**
Information Extraction – Named entity recognition, Relation detection and classification	

Temporal and event processing, Template filling :	**04 Hours**
Temporal and event processing, Template filling.	

Course Outcomes: At the end of the course student will be able to

1.	Understand the basics of NLP along with various real world applications, perform language analysis and learn the basic syntax of the English language.
2.	Learn to perform finite-state morphological parsing, detect and correct spelling errors, implement n-gram models and understand Hidden Markov Models (HMM).
3.	Implement various syntactic parsing techniques, comparing depth-first and breadth-first strategies.
4.	Comprehend feature systems and augmented grammars, developing and implementing feature-based grammars for English.
5.	Develop practical skills in morphological analysis and the integration of these techniques with the lexicon, preparing them for advanced research or professional work in NLP.

Course Outcomes Mapping with Program Outcomes & Program Specific Outcomes

Program Outcomes→	1	2	3	4	5	6	7	8	9	10	11	12	**PSO**	**PSO**
↓ Course Outcomes													1	2
22MCA352.1	3	3	3	3	-	-	3	3	-	-	-	3	3	3
22MCA352.2	3	3	3	3	-	-	3	3	-	-	-	3	3	3
22MCA352.3	3	3	3	3	-	-	3	3	-	-	-	3	3	3
22MCA352.4	3	3	3	3	-	-	3	3	-	-	-	3	3	3
22MCA352.5	3	3	3	3	-	-	3	3	-	-	-	3	3	3

1: Low 2: Medium 3: High

TEXT BOOKS:

1.	Jurafsky, D. and J. H. Martin, ”Speech and language processing: An Introduction to Natural Language Processing, Computational Linguistics, and Speech Recognition”, 2nd Edition, Prentice Hall, 2008
2.	Allen, James, “Natural Language Understanding”, 2nd Edition, Benjamin/Cumming, 1995

REFERENCE BOOKS:

1.	Steven Bird, S., Klein, E., Loper, E, “Natural Language Processing with Python-Analyzing Text with the Natural Language Toolkit”, O'Reilly Media, 2010
2.	Grant S Ingersoll, Thomas S. Morton, and Andrew L. Farris,” Taming text: how to find, organize, and manipulate it” Manning Publications Co., 2013
3.	Feldman Ronen, and James Sanger,” The text mining handbook: advanced approaches in analyzing unstructured data”, Cambridge university press,2007
4.	Christopher D Manning, and Hinrich Schütze,” Foundations of statistical natural language processing”, MIT press, 1999

E Books / MOOCs/ NPTEL

1.	http://tamingtext.com/2013/01/11/taming-text-print-copies-now-available/

MANAGEMENT INFORMATION SYSTEMS			
Course Code	**22MCA355**	**Course Type**	**PEC**
Teaching Hours/Week (L:T:P: S)	**3:0:0:0**	**Credits**	**03**
Total Teaching Hours	**40**	**CIE + SEE Marks**	**50+50**

Teaching Department: Mater of Computer Applications

Course Objectives:

1.	Describe the role of information technology and decision support systems in business
2.	Introduce the fundamental principles of computer-based information systems and design and develop an understanding of the principles and techniques used
3.	Enable students understand the various knowledge representation methods and different expert system structures
4.	Introduce the various E-business models used by organizations
5.	Enable the students to use information to assess the impact of the Internet and Internet technology on electronic commerce and electronic business

UNIT - I

Systems Engineering :	**02 Hours**
System concepts, System control, Types of systems, Handling system complexity, Classes of systems, General model of MIS, Need for system analysis	

Information and Knowledge :	**03 Hours**
Information concepts, Classification of information, Methods of data and information collection, Value of information	

Introduction of MIS :	**03 Hours**
MIS: Concept, Definition, Role of the MIS, Impact of MIS, MIS and the user, Management as a control system, MIS support to the management, Management effectiveness and MIS, Organization as system. MIS: organization effectiveness	

UNIT - II

Strategic Management of Business :	**04 Hours**
Concept of corporate planning, Essentiality of strategic planning, Development of the business strategies, Type of strategies, Short-range planning, Tools of planning, MIS: Strategic business planning	

Development of MIS :	**04 Hours**
Development of long range plans of the MIS, Ascertaining the class of information, Determining the information requirement, Development and implementation of the MIS, Management of information quality in the MIS, Organization for development of MIS, MIS development process model	

UNIT - III

Role of ICT / IT Strategies /IT Solutions :	**04 Hours**
Planning fundamentals (real world cases), Organizational planning, planning for competitive advantage (SWOT Analysis), Business models and planning. Business/IT planning, identifying business/IT strategies, Implementation Challenges, Change management, Developing business systems (real world case)	

Business Process Re-Engineering :	**04 Hours**

Introduction, Business process, Process model of the organization, Value stream model of the organization, what delay the business process, Relevance of information technology, MIS and BPR

UNIT - IV

Technology of Information Systems :	**04 Hours**

Introduction, Data processing, Transaction processing, Application processing, information system processing, TQM of information systems, Human factors & user interface, Strategic nature of IT decision, MIS choice of information technology

Decision Making and DSS :	**04 Hours**

Decision making concepts; Decision making process, Decision-making by analytical modeling, Behavioral concepts in decision making, Organizational decision-making, Decision structure, DSS components, Management reporting alternatives

UNIT - V

Electronic Business Systems :	**04 Hours**

Enterprise business system – Introduction, cross-functional enterprises applications, real world case, Functional business system, Introduction, marketing systems, sales force automation, CIM, HRM, online accounting system, Customer relationship management, ERP, Supply chain management (real world cases for the above)

Client Server Architecture and E-business Technology :	**04 Hours**

Client server architecture, implementation strategies, Introduction to E-business, model of E-business, internet and World Wide Web, Intranet/Extranet, electronic, Impact of Web on Strategic management, Web enables business management, MIS in Web environment.

Course Outcomes: At the end of the course student will be able to

1.	Define MIS and analyze its significance in business organizations
2.	Develop corporate planning and strategies using MIS
3.	Adapt SWOT Analysis and apply business
4.	Summarize organizational decision-making process and also apply decision support systems
5.	Analyze the role of various Electronic Business systems such CRM,ERP,SCM in organizations

Course Outcomes Mapping with Program Outcomes & Program Specific Outcomes

Program Outcomes→ **↓ Course Outcomes**	1	2	3	4	5	6	7	8	9	10	11	12	PSO 1	PSO 2
22MCA355.1	3	3	2	2	1	-	-	-	-	-	-	1	1	1
22MCA355.2	3	2		1	1	-	-	-	-	-	-	-	1	1
22MCA355.3	2	3	3		2	-	-	-	-	-	-	-	1	1
22MCA355.4	2	2	2	3	3	-	-	-	-	-	-	-	1	1
22MCA355.5	3	2	3	1	2	-	-	-	-	-	-	-	1	1

1: Low 2: Medium 3: High

TEXT BOOKS:

1.	Waman S Jawadekar : Management Information Systems , TataMcGraw Hill, 2009
2.	James A O'Brien and George M Marakas : Management Information Systems, 7th Edition, Tata McGraw Hill, 2006

REFERENCE BOOKS:	
1.	Ralph M Stair and George W Reynolds : Principles of Information Systems, 7th Edition, Cengage Learning, 2010
2.	Steven Alter: Information Systems - The Foundation of E-Business, 4th Edition, Pearson Education Asia, 2011
3.	Mahadeo Jaiswal and Monika Mital : Management Information System, 3rd Edition, Oxford University Press
4.	Effy Oz : Management Information Systems, 5th Edition, Cengage Learning, 2006
E Books / MOOCs/ NPTEL	
1.	NPTEL course on Management Information Systems, IIT Kharagpur

TIME SERIES ANALYSIS AND PREDICTION

Course Code	**22MCA356**	**Course Type**	**PEC**
Teaching Hours/Week (L:T:P: S)	**3:0:0:0**	**Credits**	**03**
Total Teaching Hours	**40**	**CIE + SEE Marks**	**50+50**
Prerequisite	**22MCA104**		

Teaching Department: Master of Computer Applications

Course Objectives:

1.	Understand the fundamental advantage and necessity of forecasting in various situations.
2.	Derive the properties of ARMA Models
3.	Choose an appropriate ARIMA model for a given set of data and fit the model using an appropriate package
4.	Derive the properties of ARIMA and state-space models
5.	Compute forecasts for a variety of linear methods and models

UNIT - I

Introduction : **08 Hours**

Examples of Time Series, Objectives of Time Series Analysis , Some Simple Time Series Models, Some Zero-Mean Models, Models with Trend and Seasonality, A General Approach to Time Series Modeling, Stationary Models and the Auto-correlation Function, The Sample Auto-correlation Function, A Model for the Lake Huron Data, Estimation and Elimination of Trend and Seasonal Components, Estimation and Elimination of Trend in the Absence of Seasonality, Estimation and Elimination of Both Trend and Seasonality, Testing the Estimated Noise Sequence.

Stationary Processes, Basic Properties, Linear Processes, Introduction to ARMA Processes, Properties of the Sample Mean and Autocorrelation Function, Estimation of μ, Estimation of $\gamma(\cdot)$ and $\rho(\cdot)$, Forecasting Stationary Time Series, The Durbin–Levinson.

UNIT - II

ARMA Model : **08 Hours**

ARMA(p, q) Processes, The ACF and PACF of an ARMA(p, q) Process, Calculation of the ACVF, The Autocorrelation Function, The Partial Autocorrelation Function, Forecasting ARMA Processes. Spectral Analysis, Spectral Densities, The Periodogram, Time-Invariant Linear Filters, The Spectral Density of an ARMA Process.

UNIT - III

Modelling and Forecasting with ARMA Processes : **08 Hours**

Preliminary Estimation, Yule–Walker Estimation, Burg's Algorithm, The Innovations Algorithm, The Hannan–Rissanen Algorithm, Maximum Likelihood Estimation, Diagnostic Checking, Graph, The Sample ACF of the Residuals, Tests for Randomness of the Residuals, Forecasting, Order Selection, The FPE Criterion, The AICC Criterion.
Nonstationary and Seasonal Time Series Models, ARIMA Models for Nonstationary Time Series, Identification Techniques, Unit Roots in Time Series Models.

Unit - IV

State-Space Models: **08 Hours**

State-Space Representations, The Basic Structural Model, State-Space Representation of ARIMA Models , The Kalman Recursions, Estimation For State-Space Models, State-Space Models with Missing Observations, The EM Algorithm, Generalized State-Space Models, Parameter-Driven Models, Observation-Driven Models.
Forecasting: Introduction, Minimum Mean Square Error Forecasts, computation of Forecasts, The ARIMA forecast as a Weighted Average of Previous Observations, Updating Forecasts, Eventual Forecast Functions.

Unit - V

Seasonal Time Series Models : **08 Hours**

General Concepts, Traditional Methods, Regression Method, Moving Average Method, Seasonal ARIMA Models, Empirical Examples.
Exponential smoothing methods: Introduction, First-Order Exponential smoothing, Modeling Time Series Data, Second-Order Exponential Smoothing, Higher-Order Exponential Smoothing, Forecasting, Constant Process, Linear Trend Process, Exponential Smoothing for Seasonal Data, Additive Seasonal Model, Multiplicative Seasonal Model, Exponential Smoothers and Arima Models.

Course Outcomes: At the end of the course student will be able to

1.	Differentiate between models with trend and Seasonal Components.
2.	Interpret properties of ARMA models.
3.	Adapt ARIMA model for the data.
4.	Interpret properties of State-Space Models.
5.	Implement linear models.

Course Outcomes Mapping with Program Outcomes & Program Specific Outcomes

Program Outcomes→ ↓ **Course Outcomes**	1	2	3	4	5	6	7	8	9	10	11	12	**PSO** 1	**PSO** 2
22MCA356.1	3	3	3	-	-	-	-	-	-	-	-	-	2	3
22MCA356.2	3	3	3	-	-	-	-	-	-	-	-	-	2	3
22MCA356.3	3	3	3	-	-	-	-	-	-	-	-	-	2	3
22MCA356.4	3	3	3	-	-	-	-	-	-	-	-	-	2	3
22MCA356.5	2	2	2	-	2	-	-	-	-	-	-	-	2	3

1: Low 2: Medium 3: High

TEXT BOOKS :

1. Peter J. Brockwell, Richard A. Davis “Introduction to Time Series and Forecasting “, 2nd Edition, Springer Publishers
2. William W. S. Wei, “Time Series Analysis: Univariate and Multivariate Methods “, 2nd Edition, 2006, Pearson Addison Wesley
3. Douglas C. Montgomery, Cheryl L. Jennings, “Introduction to Time Series Analysis and Forecasting”, 2nd Edition, WILEY Publishers

REFERENCE BOOKS :

1. T M J A Cooray, “Applied Time Series: Analysis and Forecasting”, 2nd Edition, 2016, Narosa Publishers.
2. Dr. Avishek Pal, Dr. PKS Prakash, “Practical Time Series Analysis”, 2nd Edition, 2017, PACKT Publishers.
3. Robert H. Shumway, David S. Stoffer, “Time Series Analysis and Its Applications: With R Examples”, 2nd Edition, 2015, Springer Publishers.
4. Chris Chatfield, “The Analysis of Time Series: An Introduction“, 6th Edition, 2016, CRC Press Publishers.

E Books / MOOCs/ NPTEL	
1.	https://nptel.ac.in/courses/103106123
2.	https://onlinecourses.nptel.ac.in/noc21_ch28/preview
3.	https://www.coursera.org/learn/practical-time-series-analysis#syllabus

ARTIFICIAL INTELLIGENCE AND MACHINE LEARNING LAB

Course Code	**22MCA303**	**Course Type**	**PCC**
Teaching Hours/Week (L: T: P: S)	**0:0:4:0**	**Credits**	**02**
Total Teaching Hours	**26**	**CIE + SEE Marks**	**50+50**
Prerequisite	**22MCA104**		

Teaching Department: Master of Computer Applications

Course Objectives:

1.	To understand and implement data preprocessing techniques.
2.	To understand and implement Supervised and Unsupervised Machine Learning Algorithms.
3.	To understand and implement Deep Learning Algorithms.
4.	To understand and implement performance evaluation techniques for the model built.

List of Experiments

1.	Write basic programs (Conditional, branching, looping, Methods and modules, classes and objects, numpy, pandas, sklearn) using Python
2.	Write Programs to demonstrate supervised and unsupervised learning
3.	Write a program to demonstrate the working of the decision tree based ID3 algorithm. Use an appropriate data set for building the decision tree and apply this knowledge to classify a new sample.
4.	Build an Artificial Neural Network by implementing the Back propagation algorithm and test the same using appropriate data sets.
5.	Write a program to implement the Naive Bayesian classifier for a sample training data set stored as a .CSV file. Compute the accuracy of the classifier, considering few test data sets.
6.	Assuming a set of documents that need to be classified, use the Naive Bayesian Classifier model to perform this task. Built-in Java classes/API can be used to write the program. Calculate the accuracy, precision, and recall for your data set.
7.	Write a program to construct a Bayesian network considering medical data. Use this model to demonstrate the diagnosis of heart patients using standard Heart Disease Data Set. You can use Java/Python ML library classes/API.
8.	Apply k-Means algorithm to cluster a set of data stored in a .CSV file. You can add Java/Python ML library classes/API in the program.
9.	Write a program to implement k-Nearest Neighbor algorithm to classify the iris data set. Print both correct and wrong predictions. Java/Python ML library classes can be used for this problem.
10.	Implement the non-parametric Locally Weighted Regression algorithm in order to fit data points. Select appropriate data set for your experiment and draw graphs

Course Outcomes: At the end of the course student will be able to

1.	Apply data preprocessing techniques, supervised and unsupervised machine learning algorithms.
2.	Apply deep learning algorithms and implement performance evaluation techniques.

Course Outcomes Mapping with Program Outcomes & Program Specific Outcomes

Program Outcomes→	1	2	3	4	5	6	7	8	9	10	11	12	**PSO1**	**PSO2**
↓ Course Outcomes													1	2
22MCA303.1	3	3	3	3	3	-	3	-	2	3	3	3	3	3
22MCA303.2	3	3	3	3	3	-	3	-	2	3	3	3	3	3

1: Low 2: Medium 3: High

REFERENCE BOOKS:	
1.	Cosma Rohilla Shalizi, Advanced Data Analysis from an Elementary Point of View, 2015
2.	Tom M Mitchell,“ Machine Learning”, 1st Edition, McGraw Hill Education, 2017.
3.	Elaine Rich, Kevin K and S B Nair, “Artificial Intelligence”, 3rd Edition, McGraw Hill Education, 2017
4.	Introduction to Machine Learning with Python - A Guide for Data Scientists (Muller Andreas)
5.	Allen B. Downey, “Think Python: How to Think Like a Computer Scientist”, 2nd Edition, Updated for Python 3, Shroff/O‘Reilly Publishers, 2016

ADVANCED WEB TECHNOLOGIES LAB			
Course Code	**22MCA304**	**Course Type**	**PCC**
Teaching Hours/Week (L: T: P: S)	**0:0:4:0**	**Credits**	**02**
Total Teaching Hours	**26**	**CIE + SEE Marks**	**50+50**
Prerequisite	**22MCA109**		

Teaching Department: Master of Computer Applications

Course Objectives:

1.	To develop an ability to design and implement static and dynamic website
2.	Use appropriate client-side or Server-side applications
3.	Develop basic programming skills using Javascript and JQuery
4.	Learn the language of web : HTML,CSS, Bootstrap, PHP, JSON

List of Experiments

1.	Develop and demonstrate XHTML document that illustrate the use of images, tables, links, formatting tools, lists, forms, frames and style sheets
2.	Develop and demonstrate HTML5 document that illustrate the use of new semantics elements, Migration, canvas, svg, input types, new form elements and attribute, google map, media(audio, video).
3.	Develop and demonstrate XHTML document that includes javascript for the following a. I/O statements b. Control statements c. Arrays d. Objects e. Functions f. Constructors g. Event Handling through… Body elements Button elements Text box and password elements h. Element accessing, positioning and moving i. Stacking and slow movements to elements j. Dragging and dropping the elements k. Element visibility
4.	Develop and Demonstrate XHTML document that includes javascript for Pattern Matching Using Regular Expressions
5.	Design a XML document to store information and display the XML document.
6.	Execute simple programs using PHP
7.	Create a Web Application using PHP
8.	Database programs using PHP
9.	Simple programs using JQuey, JSON and AJAX
10.	AJAX applications using various GUI components
11.	Implementation of "Shopping cart" application using AJAX
12.	AJAX application to keep track of user data and retrieve the session data
13.	AJAX application to verify client side and server side data

Course Outcomes: At the end of the course student will be able to

1.	Design and implement static and dynamic websites with good aesthetic sense of designing and latest technical know-how's
2.	Analyze and apply the role of languages like HTML, Bootstrap, CSS, XML, JavaScript, PHP and protocols in the workings of the web and web applications

Course Outcomes Mapping with Program Outcomes & Program Specific Outcomes

Program Outcomes→	1	2	3	4	5	6	7	8	9	10	11	12	**PSO**	**PSO**
↓Course Outcomes													1	2
22MCA304-1.1	3	3	3	3	2	-	3	3	2	3	3	-	3	-
22MCA304-1.2	3	3	2	2	3	-	3	3	3	3	2	-	3	-

1: Low 2: Medium 3: High

REFERENCE BOOKS:

1. Robert W. Sebesta: Programming with World Wide Web, 4th Edition, Pearson Education, 2008

2. M. Deitel, P.J. Deitel, A. B. Goldberg: Internet & World Wide Web, How to Program, 5th Edition, Pearson Education, 2008

E Resources

1. https://onlinecourses.swayam2.ac.in/nou22_cs03/preview

2. https://onlinecourses.swayam2.ac.in/aic20_sp32/preview

MINI PROJECT

Course Code	**22MCA305**	**Course Type**	**PCC**
Teaching Hours/Week (L:T:P: S*) ***** Self Learning	**0:0:4:9**	**Credits**	**04**
Total Teaching Hours	**48**	**CIE + SEE Marks**	**50+50**

Teaching Department: Master of Computer Applications

Course Objectives:

1.	To provide the student the experience of developing quality software solutions by involving in all the stages of the software development life cycle.
2.	To learn to work in a team and present the work effectively.

The objective of the MCA mini project work is to provide the student with the experience of developing quality software solutions by involving in all the stages of the software development life cycle like requirements engineering, systems analysis, systems design, software development, testing strategies and documentation with an overall emphasis on the development of reliable software systems. The primary emphasis of the project work is to understand and gain the knowledge of the principles of software engineering practices.
Mini project can be assigned to an individual student or to a group having not more than 3 students. The mini project has to be carried out during the third semester of the MCA program. The CIE marks awarded for the Mini-project work, shall be based on the evaluation of the work done in every stage of the software development life cycle. A student has to submit a project report at the end of the semester. Contribution to the Mini-project and the performance of each group member shall be assessed individually in the semester end examination (SEE) conducted at the department

Course Outcomes: At the end of the course student will be able to

1.	Apply the principles of software engineering practices and develop a correct software for a real world problem.
2.	Create the project and defend the work effectively.

Course Outcomes Mapping with Program Outcomes & Program Specific Outcomes

Program Outcomes→	1	2	3	4	5	6	7	8	9	10	11	12	**PSO**	**PSO**
↓ Course Outcomes													1	2
22MCA305.1	3	3	3	3	3	3	3	3	3	3	3	3	3	3
22MCA305.2	3	3	3	3	3	3	3	3	3	3	3	3	3	3

1: Low 2: Medium 3: High

REFERENCE BOOKS :

1.	Roger S. Pressman, ―Software Engineering – A Practitioner's Approach, Seventh Edition, Mc Graw-Hill International Edition, 2010
2.	Ali Bahrami, "Object oriented systems development using the unified modelling language", 1st Edition, McGraw-Hill, 1998
3.	John Sonmez, "Soft Skills: The software developer's life manual", 29 December 2014
4.	Ian Sommerville, ―Software Engineering‖, 9th Edition, Pearson Education Asia, 2011